

The transfer characteristic along with output for sinusoidal input is shown in Fig. 1.44.

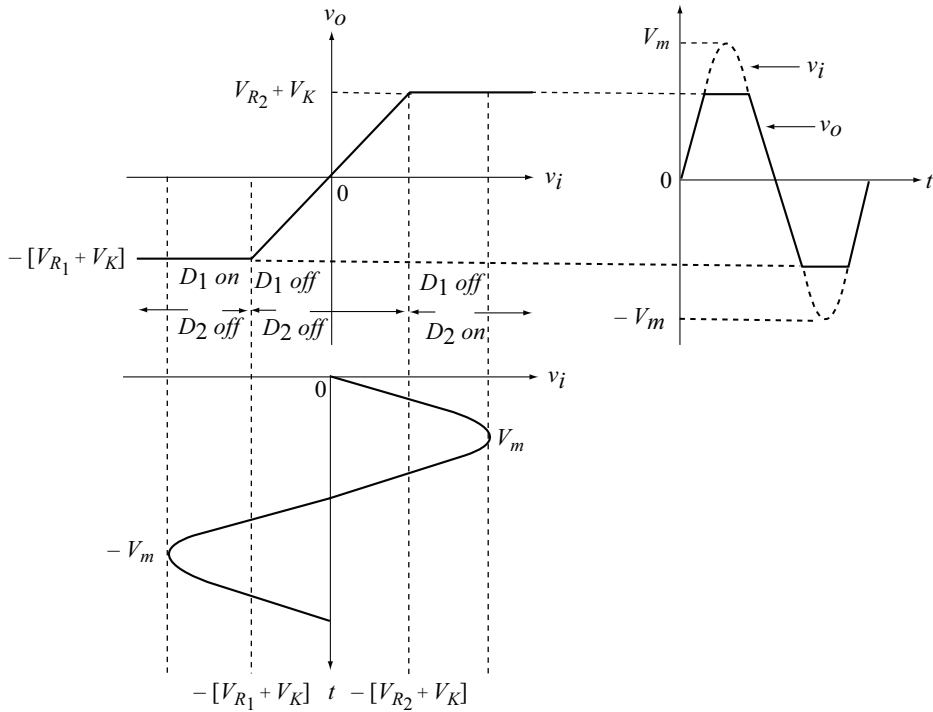


Fig. 1.44 Transfer characteristics and output waveform of the double ended clipper of Fig. 1.43

The output voltage waveforms for other inputs are shown in Fig. 1.45.

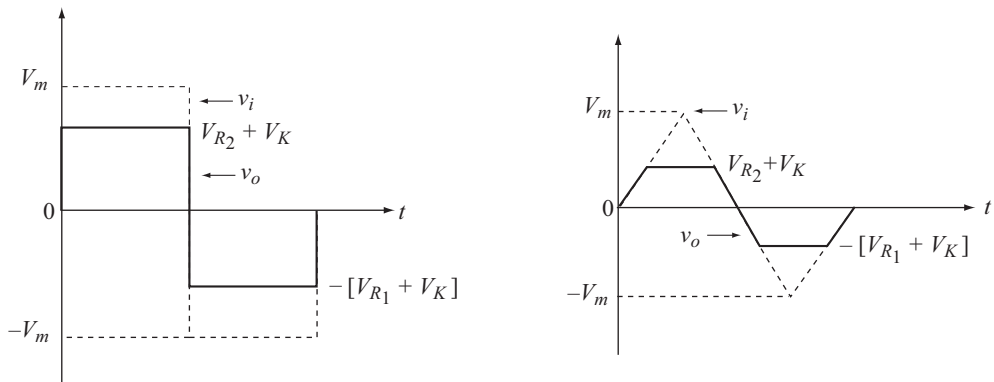
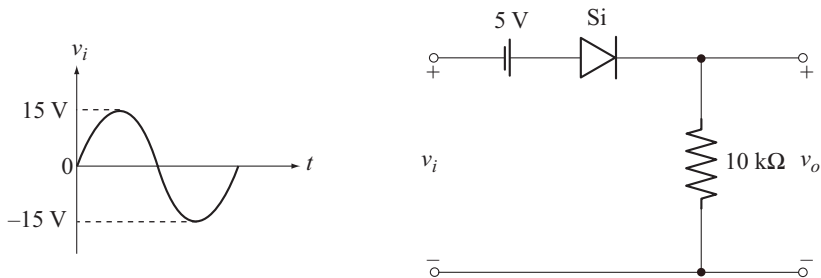


Fig. 1.45 Output waveform for square and triangular inputs

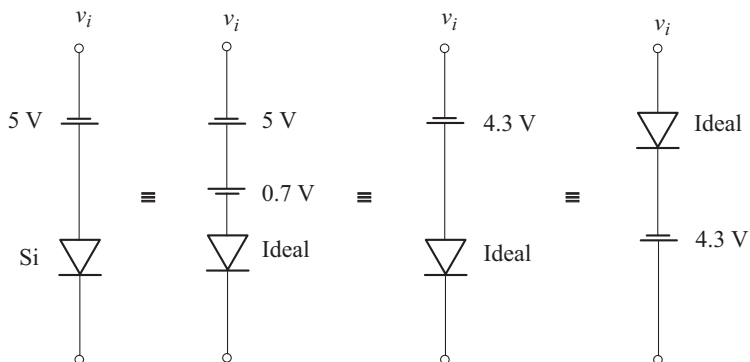
If $|V_{R1}| = |V_{R2}|$ then clipping level of the positive peak equals that of negative peak. This operation is called symmetrical clipping and the corresponding circuit is called symmetrical double ended clipper.

Example 1.32

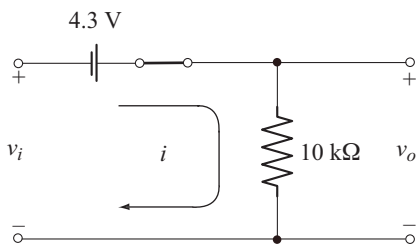
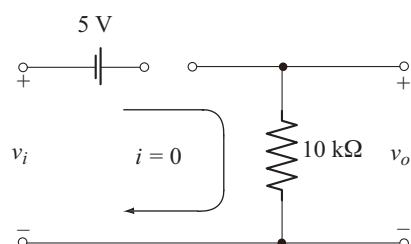
For the clipping circuit shown below, determine the transfer characteristic and sketch the output waveform.

**Solution**

First let us find the voltage on diode terminals.



Diode conducts for $v_i \geq -4.3$ V. The equivalent circuit during diode conduction is shown in Fig. A.

**Fig. A****Fig. B**

Writing KVL equation to the circuit of Fig. A.

$$\begin{aligned} v_i + 4.3 \text{ V} - v_o &= 0 \\ v_o &= v_i + 4.3 \text{ V} \quad \text{for } v_i \geq -4.3 \text{ V} \end{aligned} \quad (\text{A})$$

when

$$v_i = 15 \text{ V}$$

$$v_o = 15 \text{ V} + 4.3 \text{ V} = 19.3 \text{ V}$$

Also from Equation (A)

$$\Delta v_o = \Delta v_i$$

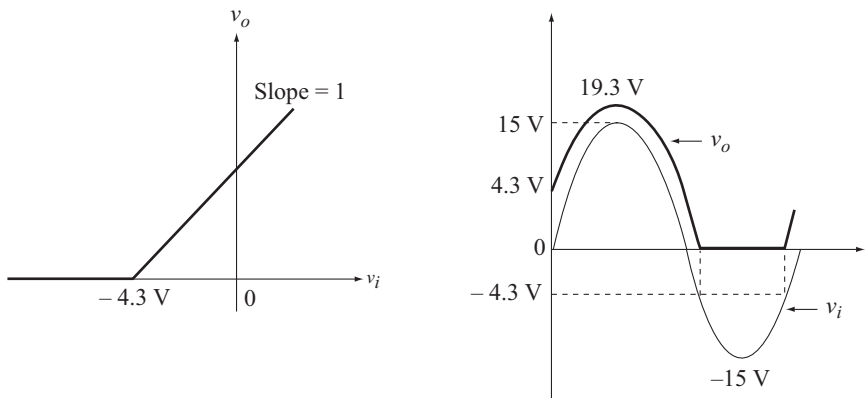
$$\Rightarrow \text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 1$$

For $v_i < -4.3 \text{ V}$, diode is off. The equivalent circuit is shown in Fig. B. From the equivalent circuit.

$$v_o = 0 \quad \text{for} \quad v_i < -4.3 \text{ V} \quad (\text{B})$$

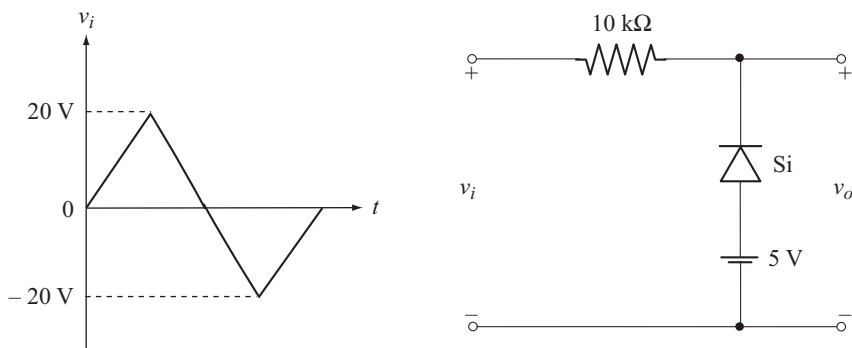
$$\text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 0$$

The transfer characteristic and output waveforms are shown below.



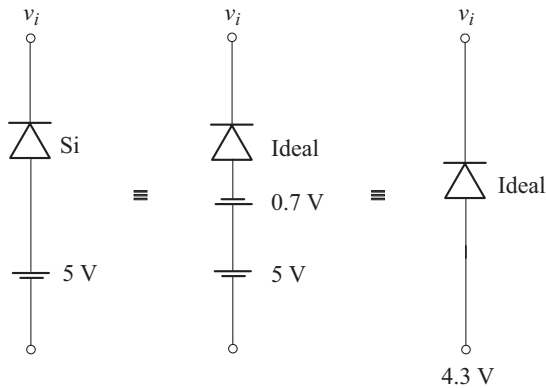
Example 1.33

For the circuit shown below determine the transfer characteristic and sketch the output waveform.

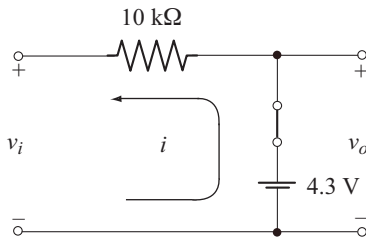
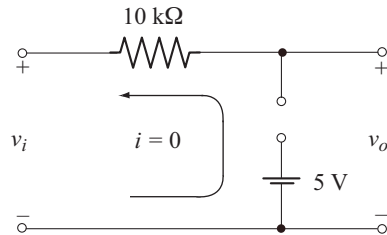


Solution

Just prior to the conduction of diode, current through $10\text{ k}\Omega$ resistor is zero. Hence v_i is available at the cathode of diode. Let us find the terminal voltages at the diode.



Diode conducts for $v_i \leq 4.3\text{ V}$. The equivalent circuit is shown in Fig. A.

**Fig. A****Fig. B**

From the equivalent circuit of Fig. A,

$$v_o = 4.3\text{ V} \quad \text{for } v_i \leq 4.3\text{ V} \quad (\text{A})$$

$$\Delta v_o = 0$$

$$\Rightarrow \text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 0$$

For $v_i > 4.3\text{ V}$, diode is off and the equivalent circuit is shown in Fig. B. Writing KVL to the circuit of Fig. B,

$$v_i + i(10\text{ k}\Omega) - v_o = 0$$

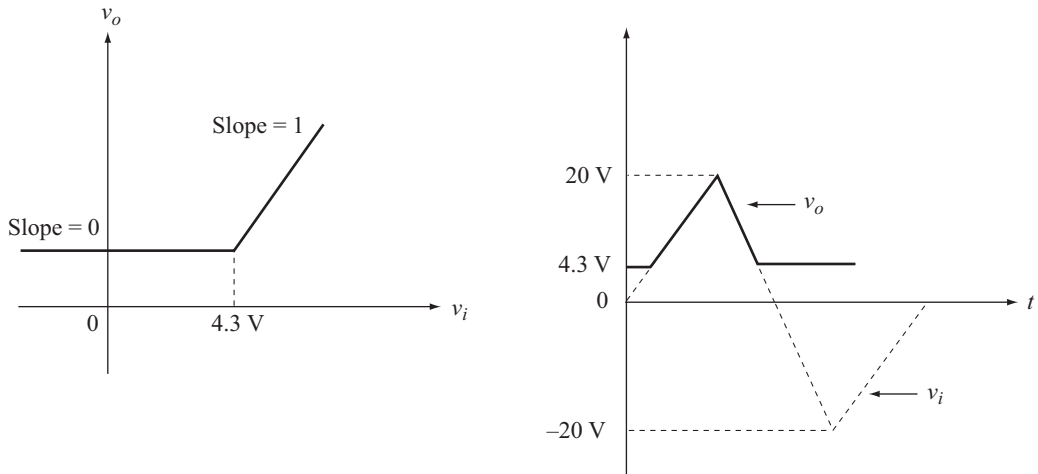
Since

$$i = 0$$

$$v_o = v_i \quad \text{for } v_i > 4.3\text{ V} \quad (\text{B})$$

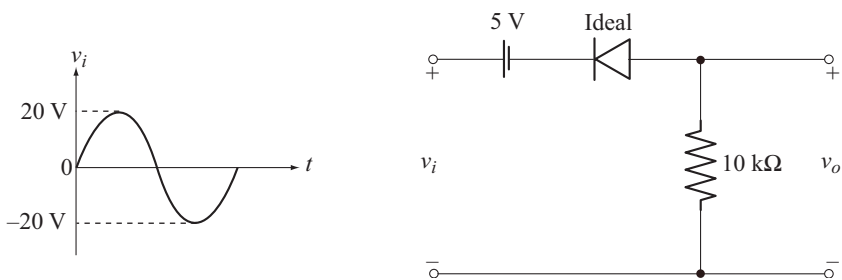
$$\text{Slope} = \frac{\Delta v_o}{\Delta v_i} = 1$$

The transfer characteristic and output waveform are shown below.



Example 1.34

For the circuit shown below determine the transfer characteristic and sketch the output waveform.



Solution

Diode conducts for $v_i \leq 5$ V. The equivalent circuit during diode conduction is shown in Fig. A.

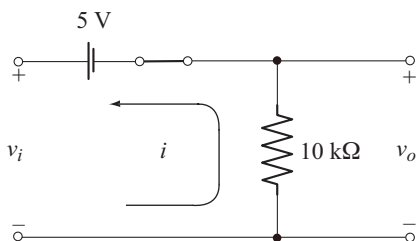


Fig. A

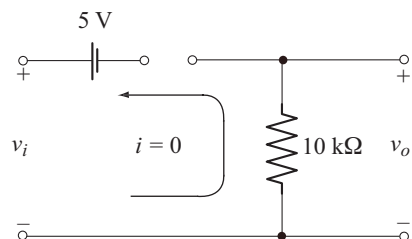


Fig. B

Applying KVL to the circuit of Fig. A, we have

$$\begin{aligned} v_i - 5 \text{ V} - v_o &= 0 \\ v_o &= v_i - 5 \text{ V} \quad \text{for } v_i \leq 5 \text{ V} \end{aligned} \quad (\text{A})$$

When $v_i = -20 \text{ V}$

$$v_o = -20 \text{ V} - 5 \text{ V} = -25 \text{ V}$$

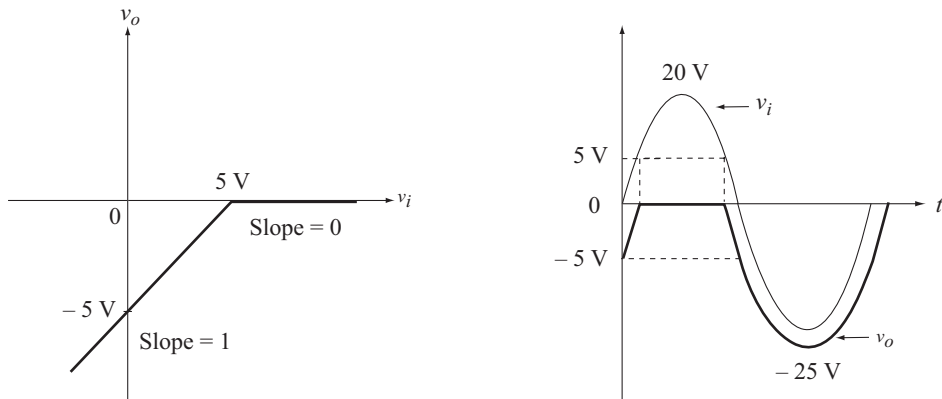
Also from Equation (A)

$$\begin{aligned} \Delta v_o &= \Delta v_i \\ \Rightarrow \text{Slope} &= 1 \end{aligned}$$

For $v_i > 5 \text{ V}$, diode is off and the equivalent circuit is shown in Fig. B. We find that

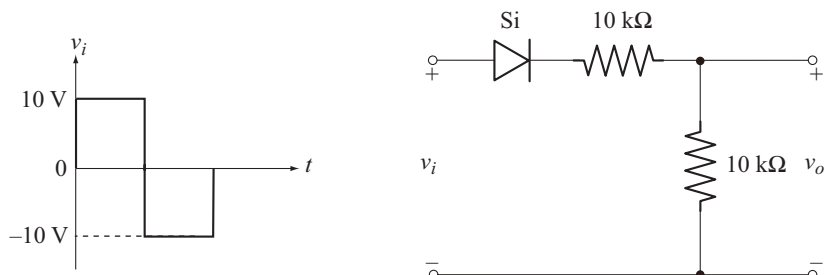
$$\begin{aligned} v_o &= 0 \quad \text{for } v_i > 5 \text{ V} \\ \Rightarrow \text{Slope} &= 0 \end{aligned} \quad (\text{B})$$

The transfer characteristic and the output waveform are shown below.



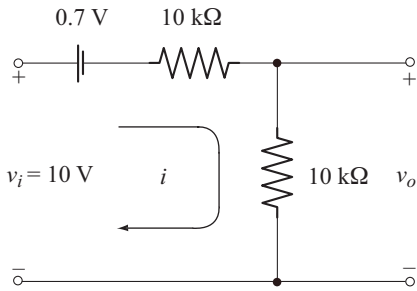
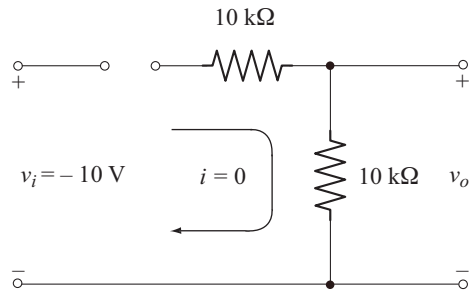
Example 1.35

For the circuit shown below sketch the output waveform for the input shown.



Solution

Diode conducts when $v_i = 10$ V. The equivalent circuit is shown in Fig. A.

**Fig. A****Fig. B**

Applying KVL to the circuit of Fig. A.

$$10 \text{ V} - 0.7 \text{ V} - i(20 \text{ k}\Omega) = 0$$

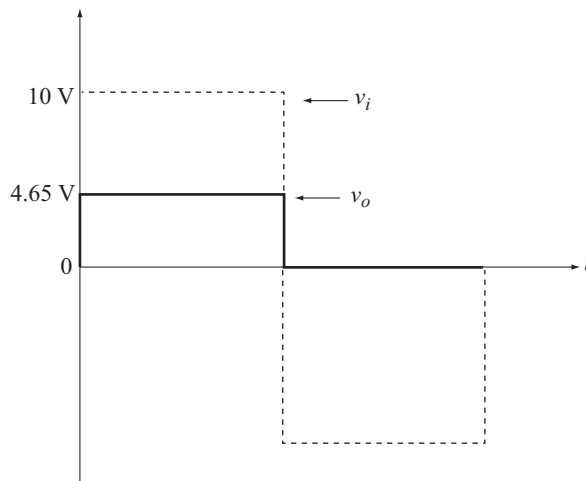
$$i = \frac{9.3 \text{ V}}{20 \text{ k}\Omega} = 0.465 \text{ mA}$$

$$v_o = i(10 \text{ k}\Omega) = 4.65 \text{ V}$$

When $v_i = -10$ V, diode turns off. The equivalent circuit is shown in Fig. B.

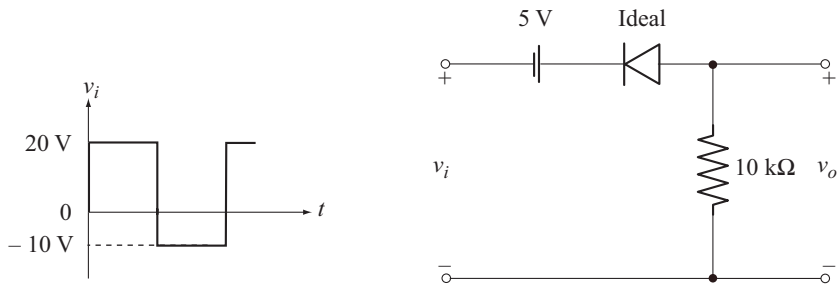
$$v_o = i(10 \text{ k}\Omega) = 0 \text{ V}$$

The waveform of v_o is shown below.

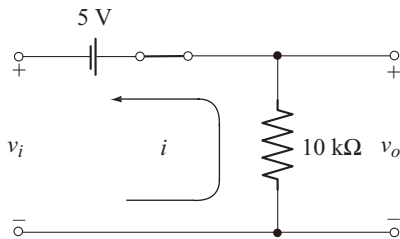
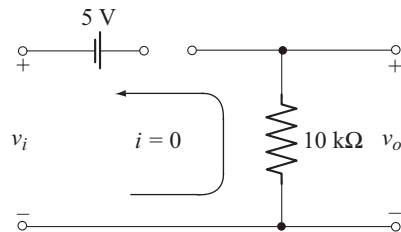


Example 1.36

For the circuit shown determine the transfer characteristic and sketch the waveform of v_o .

**Solution**

Diode conducts for $v_i \leq -5$ V. The equivalent circuit is shown in Fig. A.

**Fig. A****Fig. B**

Applying KVL to the Circuit of Fig. A we have

$$\begin{aligned}
 v_i + 5 \text{ V} - v_o &= 0 \\
 v_o &= v_i + 5 \text{ V} \quad \text{for } v_i \leq -5 \text{ V} \\
 \text{Slope} &= \frac{\Delta v_o}{\Delta v_i} = 1
 \end{aligned} \tag{A}$$

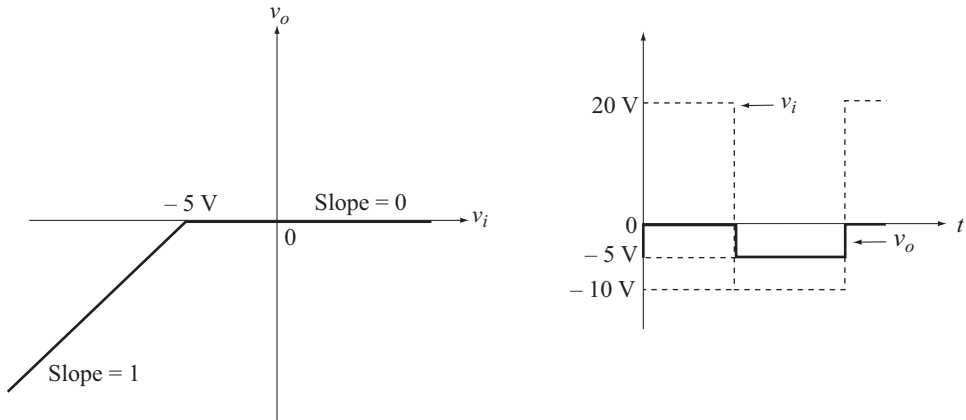
When $v_i = -10$ V,

$$v_o = -10 \text{ V} + 5 \text{ V} = -5 \text{ V}$$

For $v_i > -5$ V, diode turns off and the equivalent circuit is shown in Fig. B. Since $i = 0$,

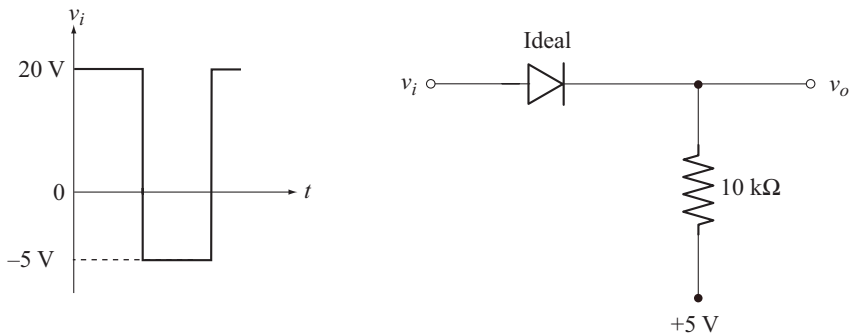
$$\begin{aligned}
 v_o &= 0 \quad \text{for } v_i > -5 \text{ V} \\
 \Rightarrow \quad \text{Slope} &= 0 \\
 \text{Also,} \quad v_o &= 0 \quad \text{when } v_i = 20 \text{ V}
 \end{aligned} \tag{B}$$

The transfer characteristic and the output waveform are shown below.



Example 1.37

For the circuit shown below determine the transfer characteristic and sketch the output voltage waveform.



Solution

Just prior to the diode conduction, no current flows through $10\text{ k}\Omega$ resistor. Hence $+5\text{ V}$ is applied on the cathode terminal and therefore diode conducts for $v_i \geq 5\text{ V}$.

The equivalent circuit is shown in Fig. A.

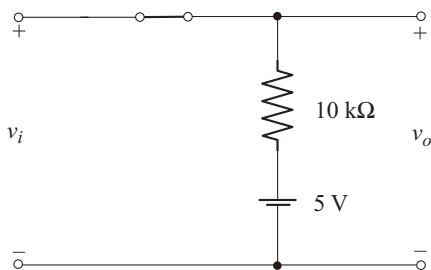


Fig. A

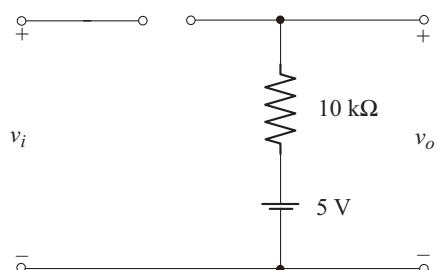


Fig. B

Applying KVL to the circuit of Fig. A, we get

$$v_i - v_o = 0$$

$$\text{or} \quad v_o = v_i, \quad \text{for } v_i \geq 5 \text{ V} \quad (\text{A})$$

$$\Rightarrow \quad \text{Slope} = 1$$

$$\text{Also, when } v_i = 20 \text{ V}, \quad v_o = 20 \text{ V}$$

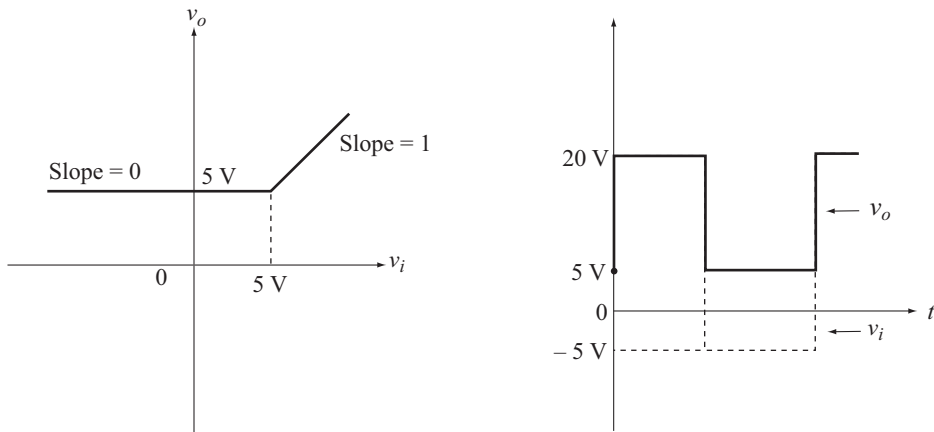
For $v_i < 5 \text{ V}$, diode turns off and the equivalent circuit is shown in Fig. B. From the equivalent circuit we find that

$$v_o = 5 \text{ V}, \quad \text{for } v_i < 5 \text{ V} \quad (\text{B})$$

$$\Rightarrow \quad \text{Slope} = 0$$

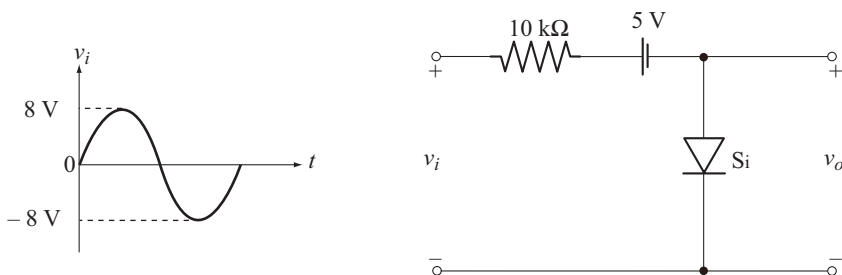
$$\text{Also when } v_i = -5 \text{ V}, \quad v_o = 5 \text{ V}$$

The transfer characteristic and output waveform are shown below.



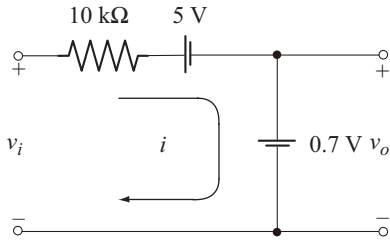
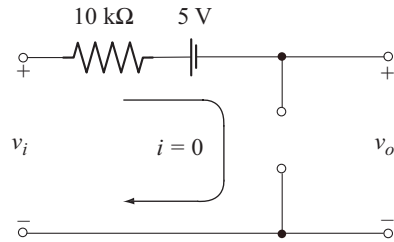
Example 1.38

For the circuit shown below sketch the output voltage waveform.



Solution

Diode conducts for $v_i \geq 5.7 \text{ V}$ and the equivalent circuit is shown in Fig. A.

**Fig. A****Fig. B**

From the circuit of Fig. A,

$$v_o = 0.7 \text{ V} \quad \text{for } v_i \geq 5.7 \text{ V} \quad (\text{A})$$

For $v_i < 5.7 \text{ V}$, diode turns off. The equivalent circuit is shown in Fig. B. Applying KVL to the circuit of Fig. B we have

$$v_i - i(10 \text{ k}\Omega) - 5 \text{ V} - v_o = 0$$

Since

$$i = 0$$

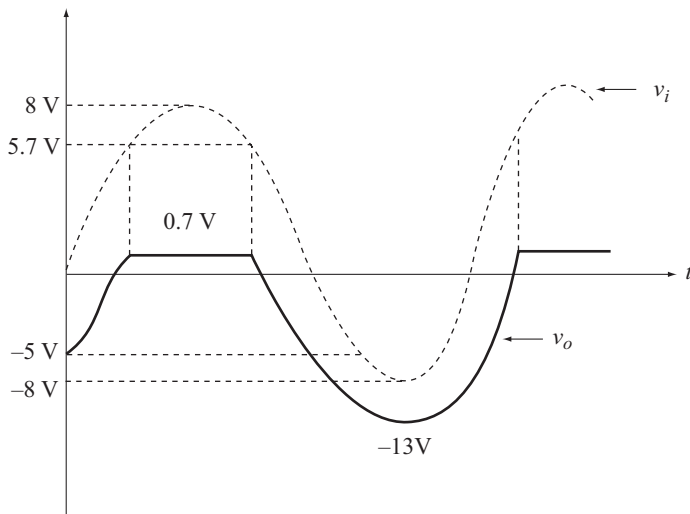
$$v_o = v_i - 5 \text{ V} \quad \text{for } v_i < 5.7 \text{ V} \quad (\text{B})$$

When

$$v_i = -8 \text{ V}$$

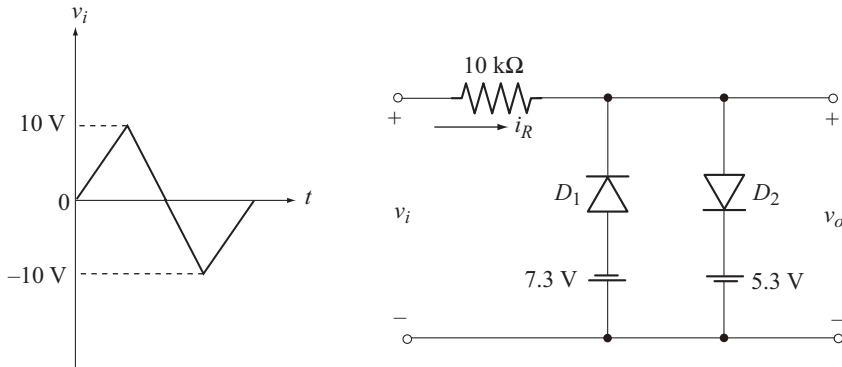
$$v_o = -8 \text{ V} - 5 \text{ V} = -13 \text{ V}$$

The output voltage waveform is shown below.



Example 1.39

For the circuit shown sketch the waveforms of i_R and v_o . Assume Si diodes.

**Solution**

The analysis of this circuit is given in section 1.22. Comparing the given circuit with that of Fig. 1.43, we have

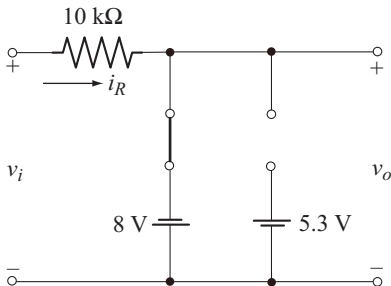
$$R = 10 \text{ k}\Omega \quad V_{R_2} = 5.3 \text{ V}$$

$$-V_{R_1} = -7.3 \text{ V} \Rightarrow V_{R_1} = 7.3 \text{ V}$$

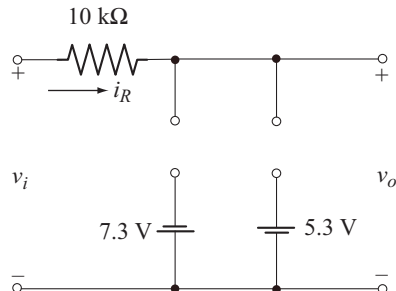
Using table 1.5, the circuit operation can be summarized as follows.

Input voltage	Diode status	Output voltage	Slope
$v_i \leq -8 \text{ V}$	D_1 on D_2 off	$v_o = -8 \text{ V}$	0
$-8 \text{ V} < v_i < 6 \text{ V}$	D_1 off D_2 off	$v_o = v_i$	1
$v_i \geq 6 \text{ V}$	D_1 off D_2 on	$v_o = 6 \text{ V}$	0

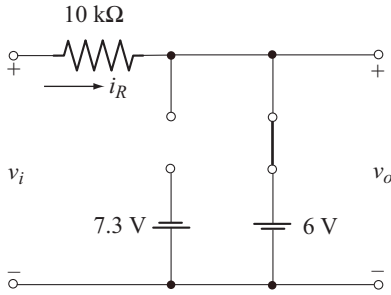
The equivalent circuits under different conditions of v_i and the transfer characteristic are shown below.



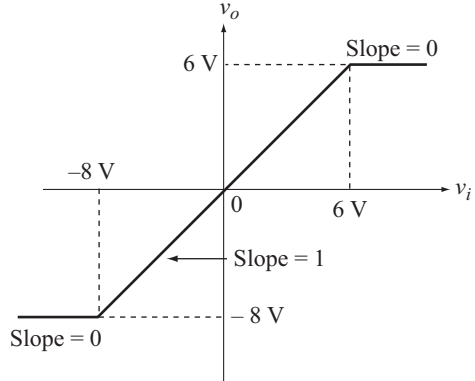
A. Equivalent circuit for
 $v_i \leq -8 \text{ V}$



B. Equivalent circuit for
 $-8 \text{ V} < v_i < 6 \text{ V}$



C. Equivalent circuit for
 $v_i \geq 6 \text{ V}$



D. Transfer characteristics

Calculation of i_R

Applying KVL to the closed path in the equivalent circuit of Fig. A, we get

$$v_i - 10 \text{ k}\Omega (i_R) + 8 \text{ V} = 0$$

$$i_R = \frac{v_i + 8 \text{ V}}{10 \text{ k}\Omega} \quad \text{for } v_i \leq -8 \text{ V} \quad (\text{A})$$

When $v_i = -10 \text{ V}$

$$i_R = \frac{-10 \text{ V} + 8 \text{ V}}{10 \text{ k}\Omega} = -0.2 \text{ mA}$$

From the equivalent circuit of Fig. B

$$i_R = 0, \quad \text{for } -8 \text{ V} < v_i < 6 \text{ V} \quad (\text{B})$$

Finally applying KVL to the closed path in the circuit of Fig. C.

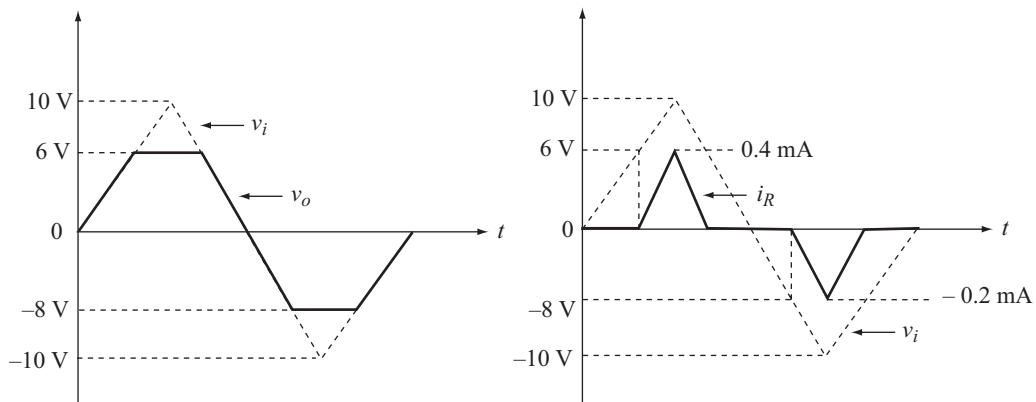
$$v_i - 10 \text{ k}\Omega (i_R) - 6 \text{ V} = 0$$

$$i_R = \frac{v_i - 6 \text{ V}}{10 \text{ k}\Omega} \quad \text{for } v_i \geq 6 \text{ V} \quad (\text{C})$$

When $v_i = 10 \text{ V}$

$$i_R = \frac{10 \text{ V} - 6 \text{ V}}{10 \text{ k}\Omega} = 0.4 \text{ mA}$$

The waveforms of v_o and i_o are shown below.

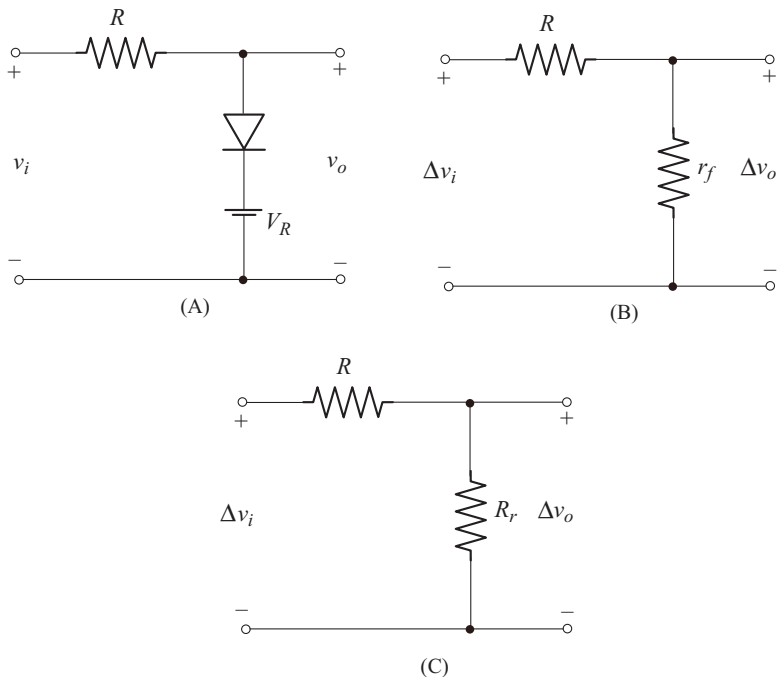


Example 1.40

Show that for a single diode clipper, a reasonable value of R is the geometric mean of r_f and R_r where r_f is the dynamic forward resistance and R_r is the reverse resistance of diode.

Solution

Let us consider the shunt clipper shown in Fig. A.



The equivalent circuit during diode conduction is shown in Fig. B. It should be noted that r_f is same as r_{av} . Since we have considered the incremental values, Δv_i and Δv_o of v_i and v_o respectively, V_K and V_R are not shown. This is because $\Delta V_K = 0$ and $\Delta V_R = 0$, since V_K and V_R are constant values.

Applying voltage division rule in the circuit of Fig. B, we get

$$\Delta v_o = \frac{r_f}{R + r_f} \Delta v_i$$

Slope of transfer characteristic is

$$\begin{aligned} \text{Slope} &= \frac{Dv_o}{Dv_i} \\ &= \frac{r_f}{R + r_f} \end{aligned}$$

For perfect clipping, slope should be zero when the diode conducts.

This requires that

$$\begin{aligned} r_f &\ll R \\ \text{Let } r_f &= \frac{R}{K} \end{aligned} \quad (A)$$

where K is some arbitrary large number.

Fig. C shows the equivalent circuit when the diode is off. The slope of transfer characteristic when the diode is off is

$$\begin{aligned} \text{Slope} &= \frac{Dv_o}{Dv_i} \\ &= \frac{R_r}{R + R_r} \end{aligned}$$

For the slope to be unity when the diode is off, we require that

$$\begin{aligned} R &\ll R_r \quad \text{or} \quad R_r \gg R \\ \text{Let } R_r &= KR \end{aligned} \quad (B)$$

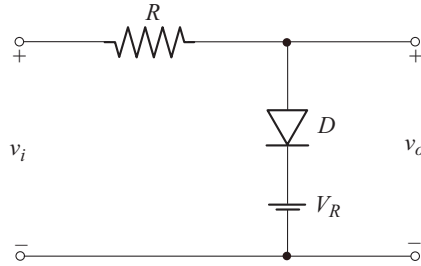
Multiplying Equations (A) and (B) we get

$$\begin{aligned} r_f R_r &= R^2 \\ \text{or } R &= \sqrt{r_f R_r} \end{aligned} \quad (C)$$

Thus a reasonable value for R would be the geometric mean of diode forward resistance r_f and the reverse resistance R_r .

Example 1.41

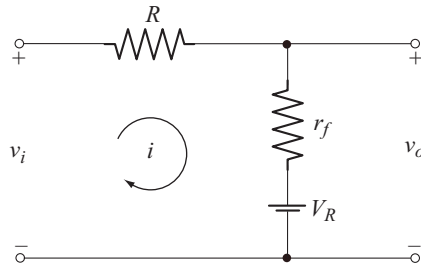
For the diode clipping circuit shown, draw the input and output waveforms for (a) $R = 100 \, \Omega$ (b) $R = 1 \, \text{k} \, \Omega$ and (c) $R = 10 \, \text{k} \, \Omega$ given that $v_i = 20 \sin \omega t$ and $V_R = 10 \, \text{V}$. Assume $r_f = 100 \, \Omega$, $R_r = \infty$ and $V_K = 0$.

**Solution:**

For $v_i \geq 10 \text{ V}$, D is on

For $v_i < 10 \text{ V}$, D is off and hence $v_o = v_i$

The equivalent circuit when D is on is shown in the following figure.



$$v_o = i r_f + V_R$$

But

$$i = \frac{v_i - V_R}{R + r_f}$$

$\therefore$

$$v_o = \frac{r_f}{R + r_f} (v_i - V_R) + V_R$$

v_o is maximum when v_i is maximum.

$\therefore$

$$v_{o_{\max}} = \frac{r_f}{R + r_f} (v_{i_{\max}} - V_R) + V_R$$

$$v_{i_{\max}} = 20 \text{ V}, \quad V_R = 10 \text{ V}, \quad r_f = 100 \Omega$$

(a) For $R = 100 \Omega$

$$v_{o_{\max}} = \frac{100 \text{ W}}{100 \text{ W} + 100 \text{ W}} (20 \text{ V} - 10 \text{ V}) + 10 \text{ V} = 15 \text{ V}$$

(b) For $R = 1 \text{ k}\Omega$

$$v_{o_{\max}} = \frac{100 \text{ W}}{1000 \text{ W} + 100 \text{ W}} (20 \text{ V} - 10 \text{ V}) + 10 \text{ V} = 10.9 \text{ V}$$

(c) For $R = 10 \text{ k}\Omega$

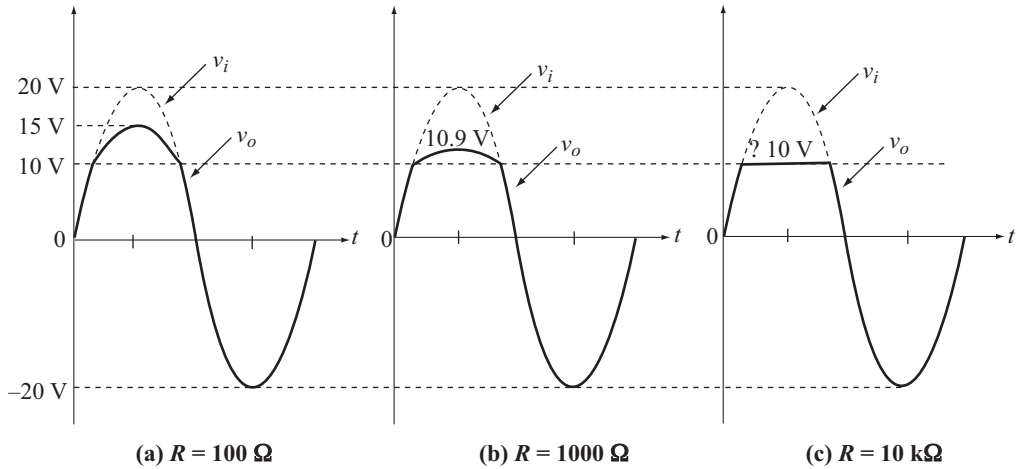
$$v_{o_{\max}} = \frac{100 \text{ W}}{10000 \text{ W} + 100 \text{ W}} (20 \text{ V} - 10 \text{ V}) + 10 \text{ V} = 10.1 \text{ V}$$

When D is off $v_o = v_i$

$\therefore$ In all the three cases

$$v_{o\min} = v_{i\min} = -20 \text{ V}$$

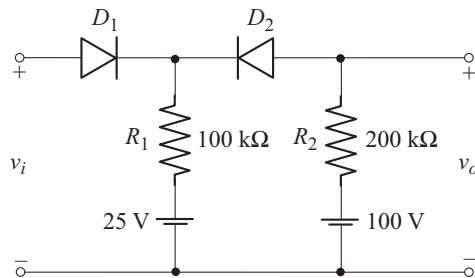
The output plots are shown below.



Note that clipping is better when $R = 10 \, \text{k}\Omega$.

Example 1.42

Sketch the output voltage waveform over the input voltage waveform for the circuit shown, given that the input varies linearly from 0 to 150 V. Assume ideal diodes.



Solution

Ideal diodes imply, $r_f = 0$, $R_r = \infty$, $V_K = 0$.

Let $v_i = 0 \text{ V}$

Then, D_1 is *off* and D_2 is *on*. The equivalent circuit is shown in Fig. A.

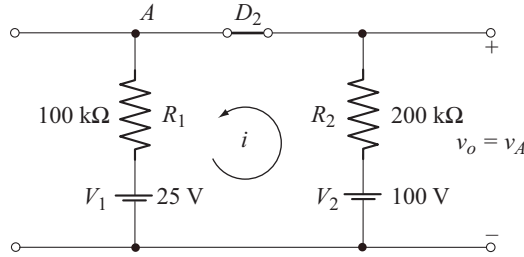


Fig. A

Applying KVL to the path of i

$$V_1 + R_1 i + R_2 i - V_2 = 0$$

$$\therefore i = \frac{V_2 - V_1}{R_1 + R_2}$$

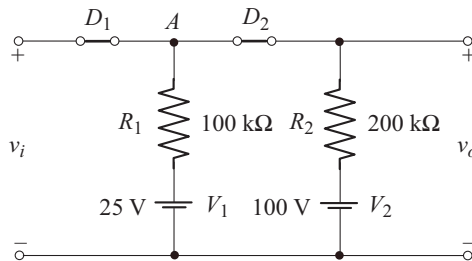
Voltage at point A

Applying KVL to the second branch

$$v_o + i R_2 - V_2 = 0$$

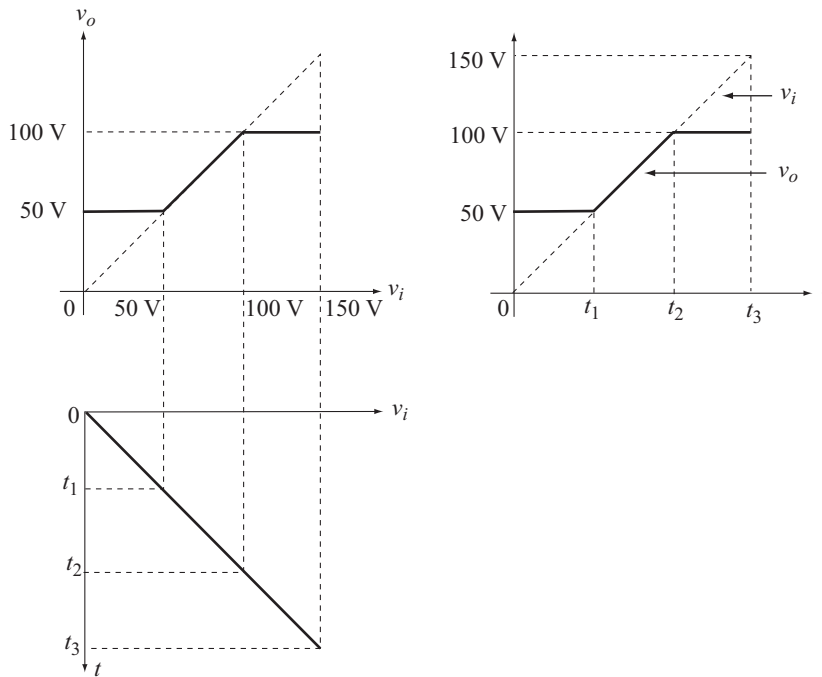
$$\begin{aligned} \therefore v_A &= v_o = -i R_2 + V_2 \\ &= \frac{-(V_2 - V_1) R_2}{R_1 + R_2} + V_2 \\ &= \frac{-(100 \text{ V} - 25 \text{ V}) 200 \text{ k}\Omega}{100 \text{ k}\Omega + 200 \text{ k}\Omega} + 100 \text{ V} \\ v_A &= 50 \text{ V} \end{aligned}$$

This situation (D_1 off, D_2 on) continues until v_i reaches 50 V, just beyond which D_1 turns on and D_2 remains on and $v_o = v_i = v_A$. The equivalent circuit is shown below.



This situation continues until v_i or v_A reaches 100 V beyond which point D_2 turns off. Hence, D_1 is on and D_2 is off and $v_o = 100 \text{ V}$.

The transfer characteristics and input-output waveforms are shown below:

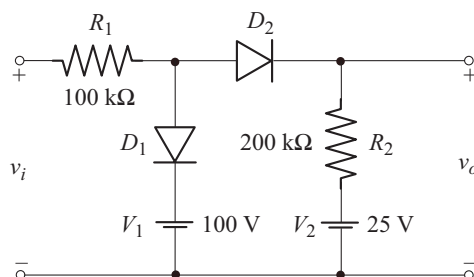


The operation of the circuit is summarised in the table given below.

Input	Output	Diode status
$v_i \leq 50$ V	$v_o = 50$ V	D_1 off, D_2 on
$50 < v_i \leq 100$ V	$v_o = v_i$	D_1 on, D_2 on
$v_i > 100$ V	$v_o = 100$ V	D_1 on, D_2 off

Example 1.43

Repeat Example 1.42 for the circuit shown.



Solution

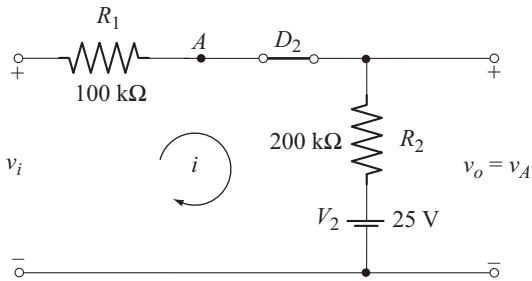
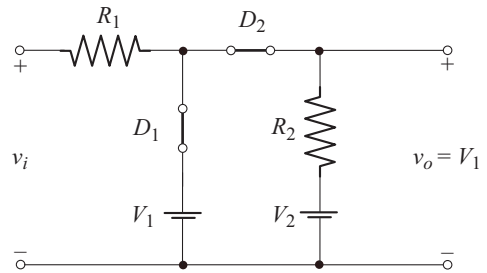
For $v_i < 25$ V, both D_1 and D_2 are off. Therefore, $v_o = 25$ V.

For v_i just exceeding 25 V, D_2 turns on and D_1 remains off. The equivalent circuit under this condition is shown in Fig. (A).

Applying KVL,

$$v_i - i(R_1 + R_2) - V_2 = 0$$

$$\therefore i = \frac{v_i - V_2}{R_1 + R_2}$$

**Fig. A****Fig. B**

$$\begin{aligned} v_o &= i R_2 + V_2 \\ &= \frac{R_2}{R_1 + R_2} (v_i - 25\text{V}) + 25 \text{ V} \\ &= \frac{2}{3} (v_i - 25\text{V}) + 25 \text{ V} \end{aligned}$$

The slope of the transfer characteristic is $\frac{2}{3}$. This situation continues until v_o or v_A reaches 100 V, beyond which D_1 also turns on. Let us use the above equation to calculate the value of v_i when $v_o = 100$ V.

$$100 \text{ V} = \frac{2}{3} (v_i - 25\text{V}) + 25 \text{ V}$$

$$\begin{aligned} \therefore v_i &= 75 \text{ V} \times \frac{3}{2} + 25 \text{ V} \\ &= 137.5 \text{ V} \end{aligned}$$

Thus, when v_i reaches 137.5 volts, v_A reaches 100 V, beyond which D_1 begins to conduct.

$\therefore$ For $v_i > 137.5$ V, D_1 on and D_2 on and $v_o = 100$ V.

The equivalent circuit is shown in Fig. (B).

These results are graphically illustrated in Fig. (C).

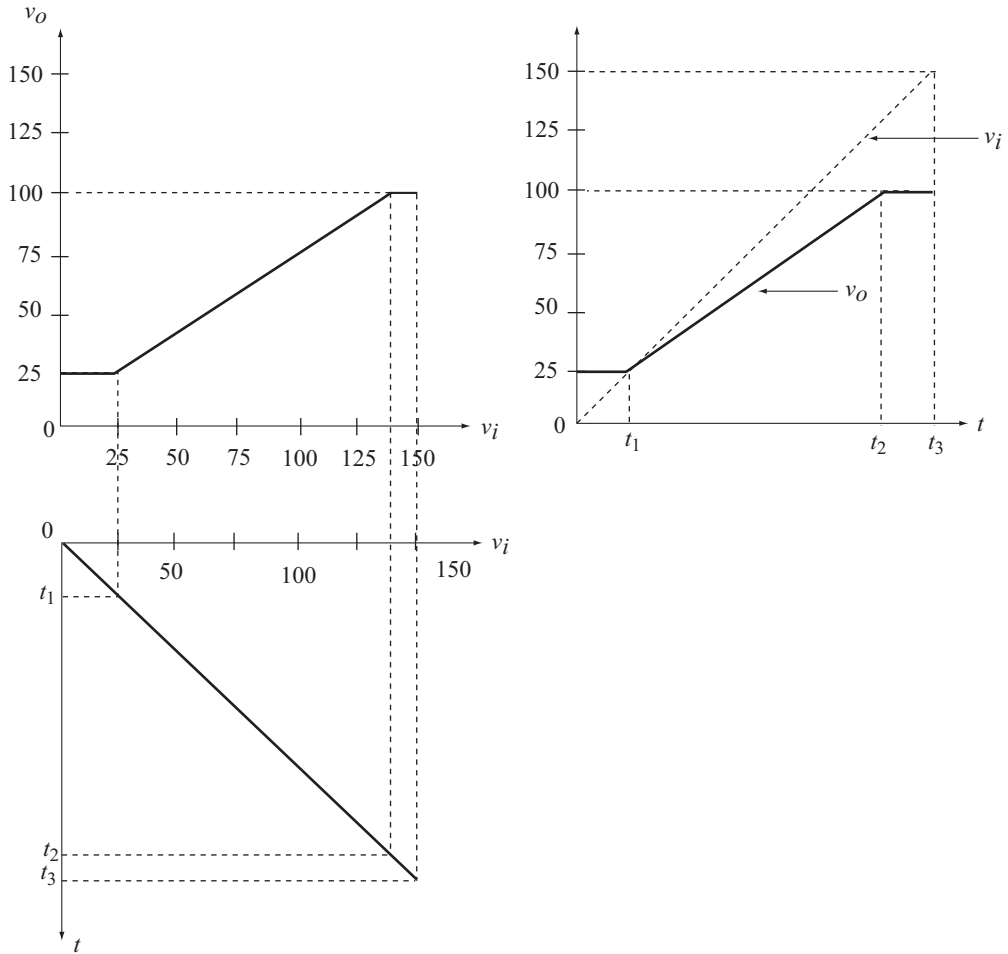


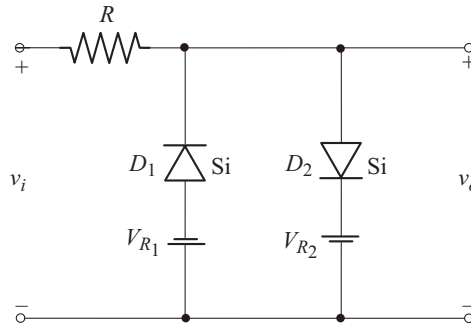
Fig. C

The circuit operation is summarised in the following table

<i>Input</i>	<i>Output</i>	<i>Slope</i>	<i>Diode status</i>
$v_i < 25 \text{ V}$	$v_o = 25 \text{ V}$	zero	D_1 off, D_2 off
$25 \text{ V} \leq v_i \leq 137.5 \text{ V}$	$v_o = \frac{2}{3} (v_i - 25 \text{ V}) + 25 \text{ V}$	slope = $\frac{2}{3}$	D_1 off, D_2 on
$v_i > 137.5 \text{ V}$	$v_o = 100 \text{ V}$	zero	D_1 on, D_2 on

Example 1.44

The circuit shown is used to square a 10 kHz sinusoidal with a peak amplitude of 50 V. Find V_{R1} and V_{R2} , if it is desired that the output voltage be flat for 90 % of the time. Assume $r_f = 100 \Omega$, $R_r = 100 \text{ k}\Omega$ for the diodes. What is the reasonable value of R ?

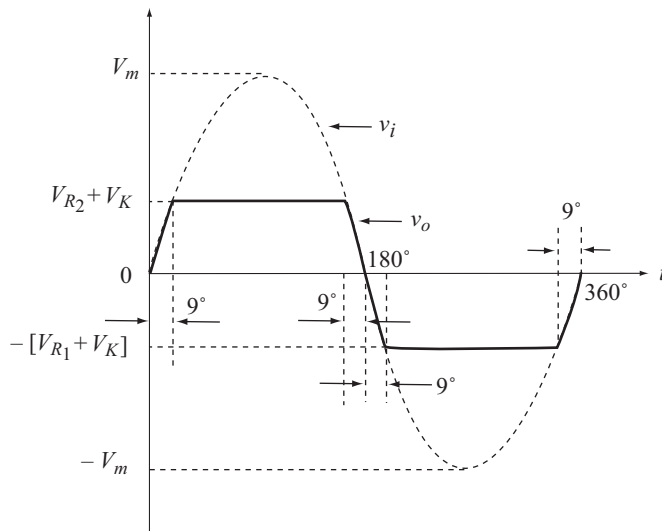
**Solution**

$$v_i = 50 \sin \theta$$

It is required that the output be flat for 90 % of the time. Now 100 % of the time corresponds to 360° and hence 90 % of the time corresponds to

$$360^\circ \times \frac{90}{100} = 324^\circ$$

In other words $v_o = v_i$ for $360^\circ - 324^\circ = 36^\circ$. This 36° is spread over four time segments as shown below:



Observe that the sinusoid becomes flat at 9° . Value of v_i at $\theta = 9^\circ$ is

$$v_i = 50 \sin 9^\circ = 7.82 \text{ V}$$

$$V_{R_1} + V_K = 7.82 \text{ V} \quad \text{and} \quad -[V_{R_2} + V_K] = -7.82 \text{ V}$$

For symmetrical clipping,

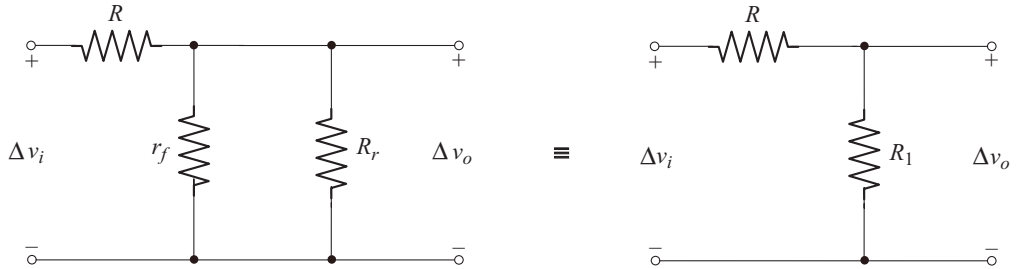
$$V_{R_1} = V_{R_2}$$

$\therefore$

$$V_{R_1} = V_{R_2} = 7.82 \text{ V} - 0.7 \text{ V} = 7.12 \text{ V}$$

To find the value of R

Consider the equivalent circuit when D_1 is on and D_2 is off. Replace D_1 by r_f and D_2 by R_r .



$$R_1 = r_f \parallel R_r = 0.1 \text{ k}\Omega \parallel 100 \text{ k}\Omega = 0.099 \text{ k}\Omega$$

$$\text{Slope} = \frac{Dv_o}{Dv_i} = \frac{R_1}{R + R_1}$$

When either of the diodes conduct, slope should be zero. This requires that,

$$R \gg R_1$$

Let

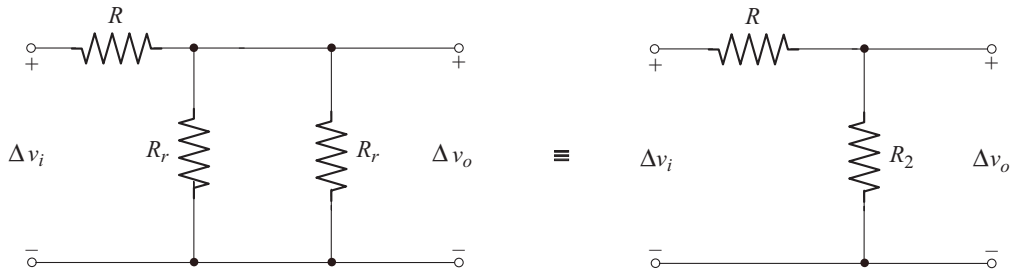
$$R = K R_1$$

or

$$K = \frac{R}{R_1}$$

(A)

Now let us consider the equivalent circuit when both diodes are off.



$$R_2 = R_r \parallel R_r = \frac{R_r}{2} = 50 \text{ k}\Omega$$

$$\text{Slope} = \frac{Dv_o}{Dv_i} = \frac{R_2}{R + R_2}$$

When both the diodes are off, slope should be unity. This requires that

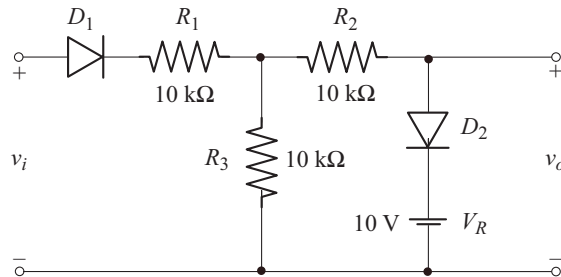
$$\begin{aligned}
 R_2 &\gg R \\
 \text{Let } R_2 &= K R \\
 \text{or } K &= \frac{R_2}{R} \quad (B)
 \end{aligned}$$

Combining Equations (A) and (B)

$$\begin{aligned}
 \frac{R}{R_1} &= \frac{R_2}{R} \\
 \text{which gives } R &= \sqrt{R_2 \times R_1} \\
 &= \sqrt{50 \text{ k}\Omega \times 0.099 \text{ k}\Omega} \\
 &= 2.22 \text{ k}\Omega
 \end{aligned}$$

Example 1.45

Plot the transfer characteristic of the circuit shown assuming ideal diodes and write the transfer characteristic equations. Sketch v_o if $v_i = 40 \sin \omega t$.



Solution

For $v_i < 0$, both D_1 and D_2 are *off*. Therefore, $v_o = 0$.

For $v_i \geq 0$, D_1 is *on*, D_2 is *off* and the equivalent circuit is shown in Fig. (A).

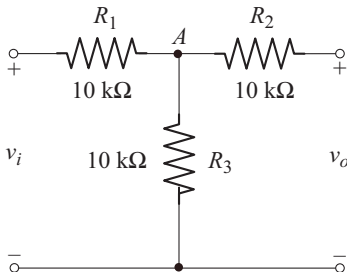


Fig. A

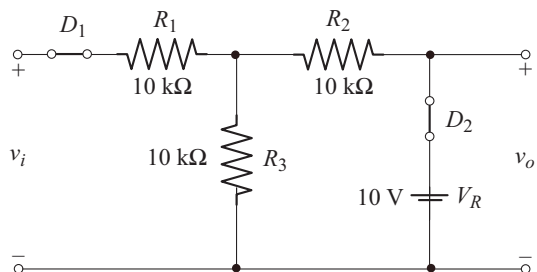


Fig. B

Since the diodes are ideal, $r_f = 0$, $R_r = \infty$ and $V_K = 0$.

$$v_o = v_A = \frac{R_3}{R_1 + R_3} v_i = \frac{v_i}{2}$$

$$\therefore \frac{v_o}{v_i} = \frac{1}{2} \quad (\text{A})$$

This situation continues until v_o reaches 10 V beyond which D_2 begins to conduct. Let us now compute v_i when v_o reaches 10 V. From Equation (A)

$$v_o = \frac{v_i}{2}$$

$$\therefore v_i = 2 v_o = 2 \times 10 \text{ V} = 20 \text{ V}$$

For $v_i \geq 20$ volts, D_2 begins to conduct and $\therefore v_o = 10$ V. The equivalent circuit is shown in Fig. (B).

The transfer equations are,
$$v_o = \begin{cases} 0 & \text{for } v_i < 0 \\ \frac{v_i}{2} & \text{for } 0 \leq v_i < 20 \text{ V} \\ 10 \text{ V} & \text{for } v_i \geq 20 \text{ V} \end{cases}$$

The transfer characteristic is shown in Fig. (C) and the output waveform is shown in Fig. (D) for $v_i = 40 \sin \omega t$.

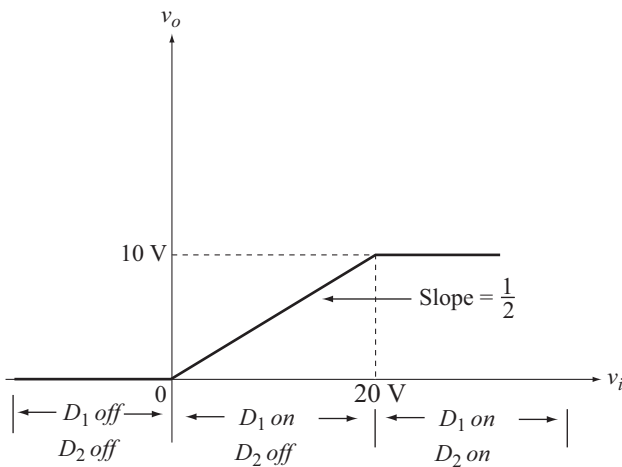


Fig. C

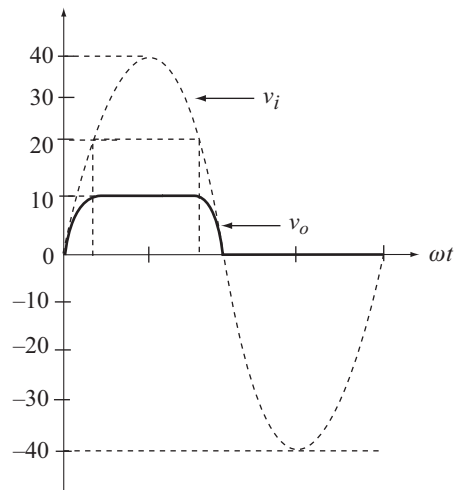
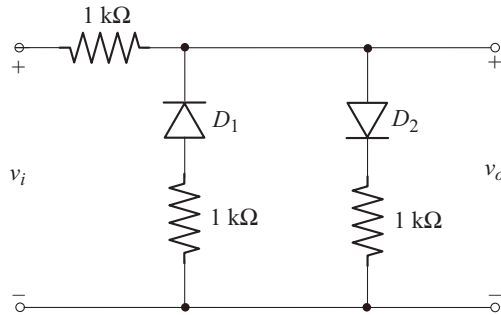


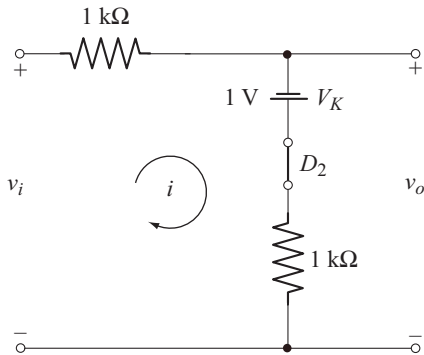
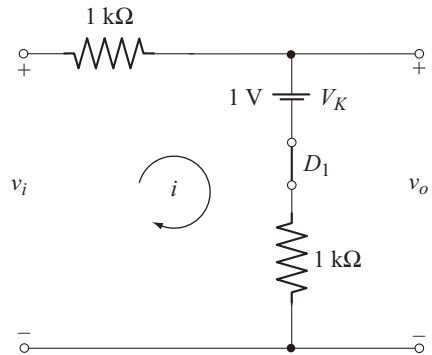
Fig. D

Example 1.46

Plot the transfer characteristic for the circuit shown and write the transfer characteristic equations. Sketch v_o if $v_i = 40 \sin \omega t$. Assume $V_K = 1$ V for the diodes.

**Solution**

For $v_i \leq -1$ V, D_1 is *on*, D_2 is *off* and the equivalent circuit is shown in Fig. (A).

**Fig. A****Fig. B**

Applying KVL to the circuit of Fig. A.

$$v_i - i(1 \text{ k}\Omega) + V_K - i(1 \text{ k}\Omega) = 0$$

$$\therefore i = \frac{v_i + V_K}{2 \text{ k}\Omega}$$

$$\text{Also } v_o + V_K - i(1 \text{ k}\Omega) = 0$$

$$v_o = -V_K + i(1 \text{ k}\Omega)$$

$$= -V_K + (v_i + V_K) \left(\frac{1 \text{ k}\Omega}{2 \text{ k}\Omega} \right)$$

$$v_o = \frac{1}{2} (v_i - V_K) \quad (\text{A})$$

For $v_i \geq 1 \text{ V}$, D_1 is off, D_2 is on and the equivalent circuit is shown Fig. (B).

$$v_o = V_K + i (1 \text{ k}\Omega)$$

But

$$i = \frac{v_i - V_K}{2 \text{ k}\Omega}$$

We get

$$v_o = \frac{1}{2} (v_i + V_K) \quad (\text{B})$$

For $-1 \text{ V} < v_i < 1 \text{ V}$, both D_1 and D_2 are off. Therefore,

$$v_o = v_i \quad (\text{C})$$

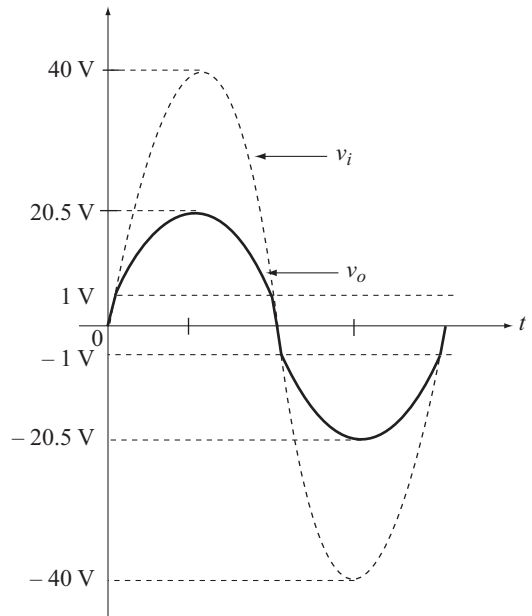
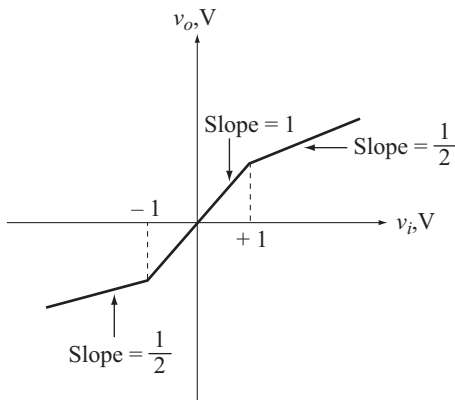
The transfer characteristic equations are

$$v_o = \begin{cases} \frac{1}{2}(v_i - 1 \text{ V}) & \text{for } v_i \leq -1 \text{ V} \\ v_i & \text{for } -1 \text{ V} < v_i < 1 \text{ V} \\ \frac{1}{2}(v_i + 1 \text{ V}) & \text{for } v_i \geq 1 \text{ V} \end{cases}$$

$$\therefore \text{From Equation (A), } v_{o\min} = \frac{1}{2} (v_{\min} - 1) = \frac{1}{2} (-40 - 1) = -20.5 \text{ V}$$

$$\therefore \text{From Equation (B), } v_{o\max} = \frac{1}{2} (v_{\max} + 1) = \frac{1}{2} (40 + 1) = 20.5 \text{ V}$$

The transfer characteristic and waveform of v_o are shown below.



Example 1.47

Repeat example 1.46 taking $V_K = 0$ V for the diodes.

Solution

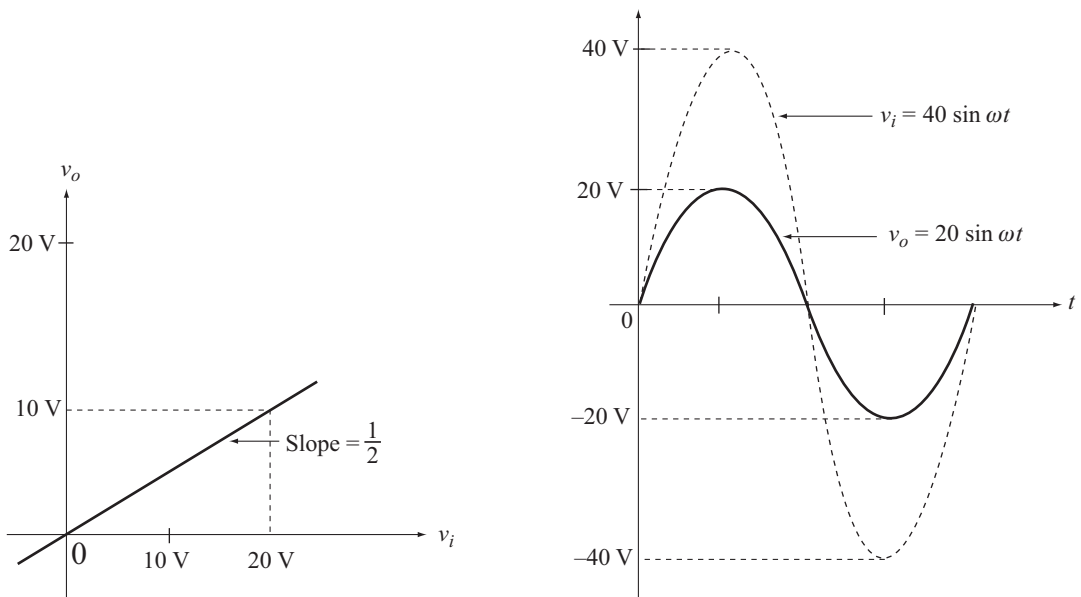
Since $V_K = 0$ V, from Equations (A), (B) and (C) of example 1.46, we find that

$$v_o = \frac{1}{2} v_i \quad \text{for all } v_i$$

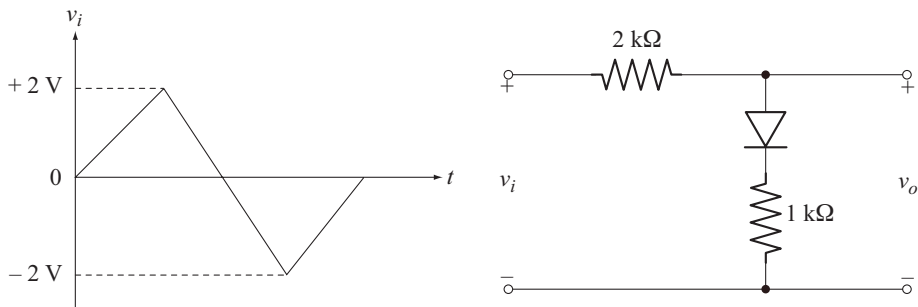
Since $v_i = 40 \sin \omega t$ V

We get $v_o = 20 \sin \omega t$ V for all v_i

The transfer characteristic and waveform of v_o are shown below.

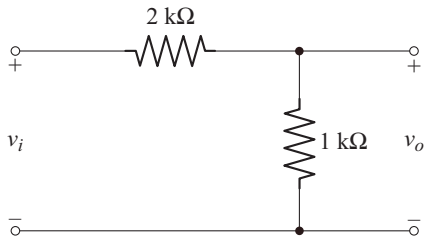
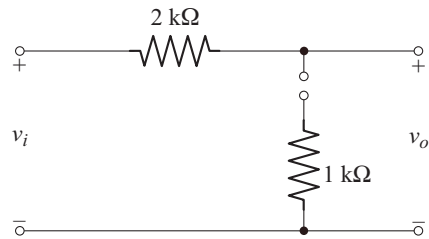
**Example 1.48**

Assuming an ideal diode in the circuit given below, draw the the output voltage waveform for the input signal given.



Solution

For $v_i \geq 0$, diode conducts. The equivalent circuit is shown in Fig. A.

**Fig. A****Fig. B**

$$v_o = \left[\frac{v_i}{2 \text{ k}\Omega + 1 \text{ k}\Omega} \right] (1 \text{ k}\Omega) = \frac{1}{3} v_i$$

$$v_{o\max} = \frac{1}{3} v_{i\max}$$

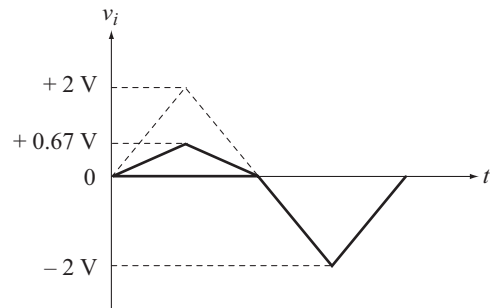
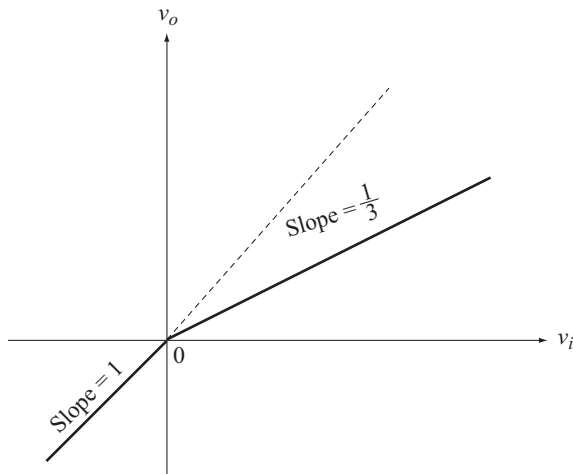
$$v_{i\max} = 2 \text{ V}$$

$$\therefore v_{o\max} = \frac{1}{3} (2 \text{ V}) = 0.67 \text{ V}.$$

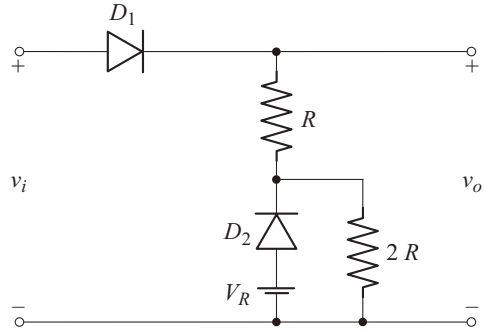
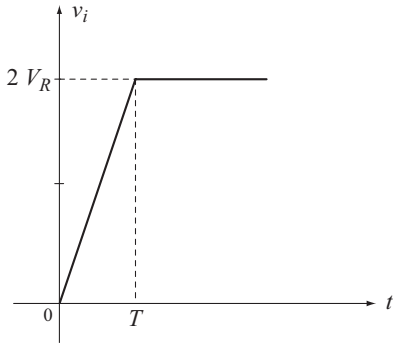
For $v_i < 0$, diode turns off. The equivalent circuit is shown in Fig. B.

From the equivalent circuit we find that $v_o = v_i$.

The transfer characteristic and the output waveform are shown below.

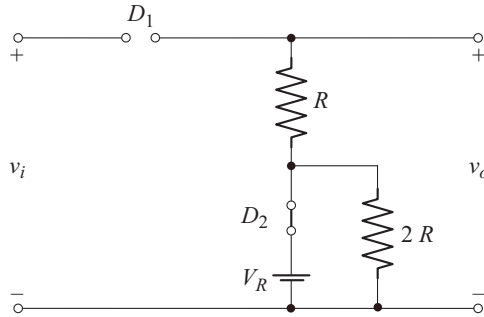
**Example 1.49**

For the circuit shown below sketch the transfer characteristic and output waveform. Assume ideal diodes.



Solution

When $v_i = 0$, D_1 is off and D_2 is on. The equivalent circuit is shown below. This situation continues as long as $v_i < V_R$.



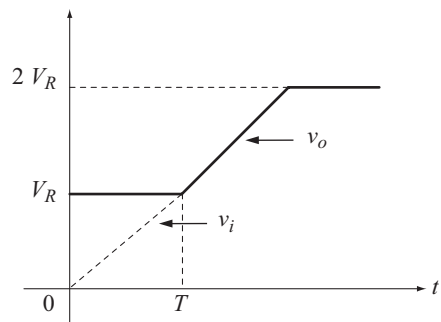
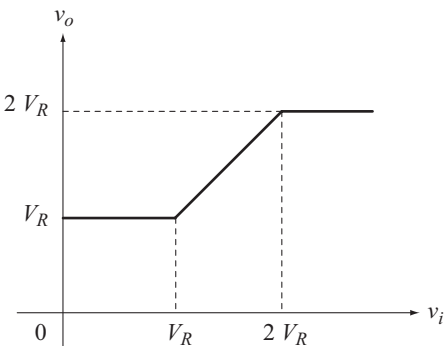
From the equivalent circuit

$$v_o = V_R \quad \text{for } v_i < V_R$$

D_1 conducts when v_i exceeds V_R . Under this condition, $v_o = v_i$

$$\therefore v_o = \begin{cases} V_R & \text{for } 0 < v_i < V_R \\ v_i & \text{for } v_i \geq V_R \end{cases}$$

The transfer characteristics and output waveform are shown below.



Example 1.50

Sketch and explain the circuit of a double ended clipper using Si diodes which limits the output between ± 10 V.

Solution:

The operation of double ended clipper which limits both positive and negative peaks is given in section 1.22.

Since clipping is symmetrical, $V_{R_1} = V_{R_2}$

$$\begin{aligned} \therefore \quad V_{R_2} + V_K &= 10 \text{ V} \quad \text{and} \quad -[V_{R_1} + V_K] = -10 \text{ V} \\ \Rightarrow \quad V_{R_1} &= V_{R_2} = 9.3 \text{ V} \end{aligned}$$

Example 1.51

Sketch and explain the circuit of a double ended clipper using Si diodes which limits the output between 5 V and 8 V.

Solution

The operation of double ended clipper which limits the output between two positive levels is given in section 1.21.

$$\begin{aligned} V_{R_1} - V_K &= 5 \text{ V} \\ \Rightarrow \quad V_{R_1} &= 5 \text{ V} + V_K = 5.7 \text{ V} \quad \text{and} \\ V_{R_2} + V_K &= 8 \text{ V} \Rightarrow V_{R_2} = 7.3 \text{ V} \end{aligned}$$

◆ 1.23 CLAMPING CIRCUITS

Clamping circuits are used to add dc level to the input signal. Clamping circuits are also called dc inserters or dc restorers. Clamping circuit uses diode, resistor and capacitor.

◆ 1.24 NEGATIVE CLAMPER

Figure 1.46 shows the circuit of negative clamper.

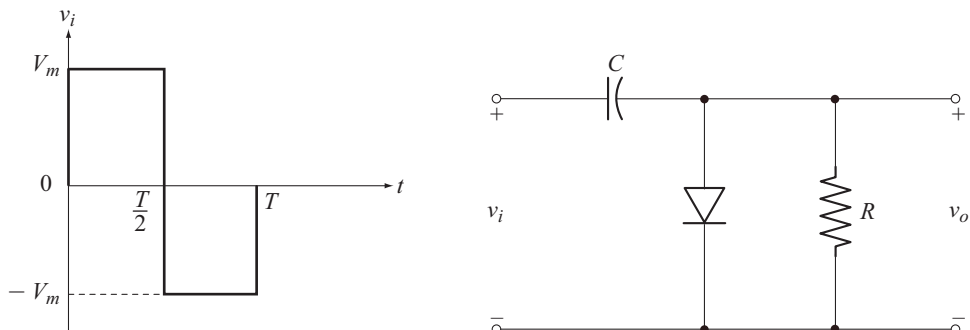


Fig. 1.46 Negative clamper

The input is a square wave which swings between $\pm V_m$ with a period $T = \frac{1}{f}$. The input can also be any other periodic waveform such as sine, triangular etc.

During the positive half cycle of v_i , the diode conducts and charges the capacitor. The charging time constant is

$$\tau_f = r_f C \quad (1.74)$$

Since the diode forward resistance is very small ($r_f \approx 0$), $\tau_f \ll \frac{T}{2}$ and hence the capacitor gets charged quickly to the peak value V_m of the input signal v_i . The equivalent circuit during diode conduction is shown in Fig. 1.47(a).

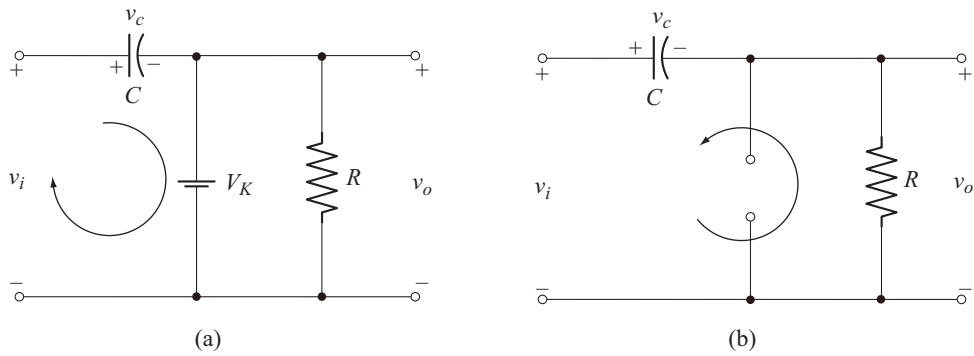


Fig. 1.47 Equivalent circuit
(a) When diode is on
(b) When diode is off

Writing KVL equation to the circuit of Fig. 1.47(a), we have

$$\begin{aligned} v_i - v_c - V_K &= 0 \\ v_c &= v_i - V_K \end{aligned} \quad (1.75)$$

When

$$\begin{aligned} v_i &= V_m \\ v_c &= V_m - V_K \end{aligned} \quad (1.76)$$

Note that the left plate of the capacitor is positive since the charging current enters that plate.

During the negative half cycle of v_i , the diode turns off. Now the capacitor discharges into R . The discharge time constant is

$$\tau = RC \quad (1.77)$$

The capacitor should not lose much of charge during discharge. To meet this requirement, the values of R and C are selected such that

$$5\tau \gg \frac{T}{2} \quad (1.78)$$

$$5 RC \gg \frac{T}{2} \quad (1.79)$$

The value of five times the time constant ($5 RC$) is made much greater than half period ($T/2$) of the periodic input signal.

The equivalent circuit during capacitor discharge is shown in Fig. 1.47(b). Writing KVL equation to the circuit of Fig. 1.47(b) we have

$$\begin{aligned} v_i - v_c - v_o &= 0 \\ v_o &= v_i - v_c \end{aligned} \quad (1.80)$$

Since

$$\begin{aligned} v_c &= V_m - V_K \\ v_o &= v_i - [V_m - V_K] \end{aligned} \quad (1.81)$$

Observe from Equation (1.81) that, the circuit adds a dc level of $-[V_m - V_K]$ to the input signal v_i . The output voltage levels corresponding to different input voltage levels are listed in Table 1.6.

Table 1.6 Output voltage levels of negative clamper

Input voltage level v_i	Output voltage level	
	Practical diode $v_o = v_i - [V_m - V_K]$	Ideal diode $v_o = v_i - V_m$
0	$-[V_m - V_K]$	$-V_m$
V_m	V_K	0
$-V_m$	$-2V_m + V_K$	$-2V_m$

The output voltage waveform is shown in Fig. 1.48.

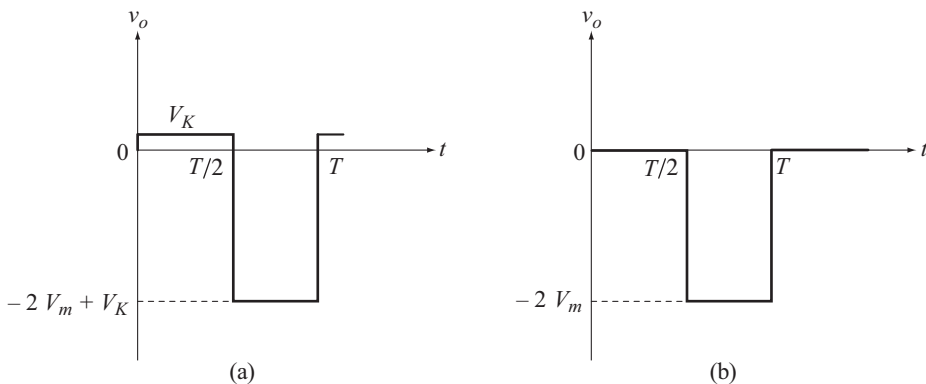


Fig. 1.48 Output voltage waveform
(a) With practical diode
(b) With ideal diode

Since the circuit adds negative dc level, it is called negative clamper. The circuit is also called positive peak clamper since the positive peak of the applied signal is clamped at 0 V as shown in Fig. 1.48(b). Also from Fig. 1.48 we observe that

peak-to-peak input voltage = peak-to-peak output voltage = $2 V_m$

Therefore in a clamper circuit

peak-to-peak output voltage is equal to peak-to-peak input voltage.

◆ 1.25 POSITIVE CLAMPER

Figure 1.49 shows the circuit of positive clamper which is obtained by simply reversing the diode in the negative clamper circuit of Fig. 1.46.

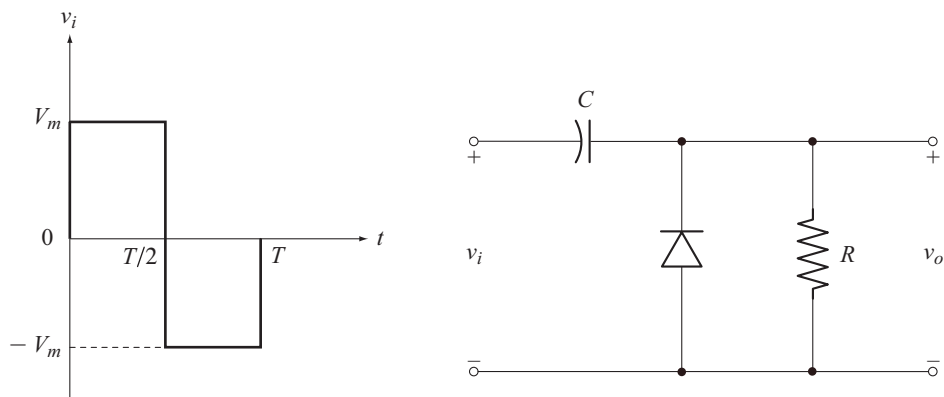


Fig. 1.49 Positive clamper

In this circuit the diode conducts during the negative half cycle of v_i and charges the capacitor to the peak value V_m , with right plate positive. Performing the analysis as outlined in the previous section, it can be shown that the circuit of Fig. 1.49 adds a dc voltage of approximately V_m to the input signal. The output voltage levels are listed in Table 1.7.

Table 1.7 Output voltage levels of positive clamper

Input voltage level v_i	Output voltage level	
	Practical diode $v_o = v_i + [V_m - V_K]$	Ideal diode $v_o = v_i + V_m$
0	$V_m - V_K$	V_m
V_m	$2V_m - V_K$	$2 V_m$
$- V_m$	$- V_K$	0

The output voltage waveform is shown in Fig. 1.50.

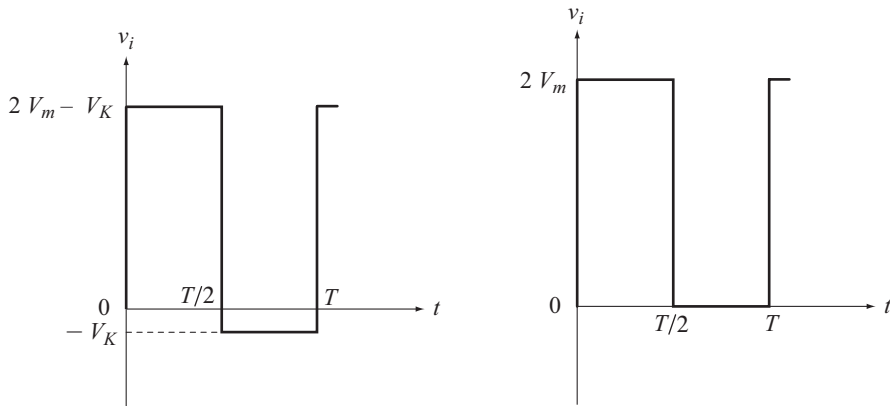


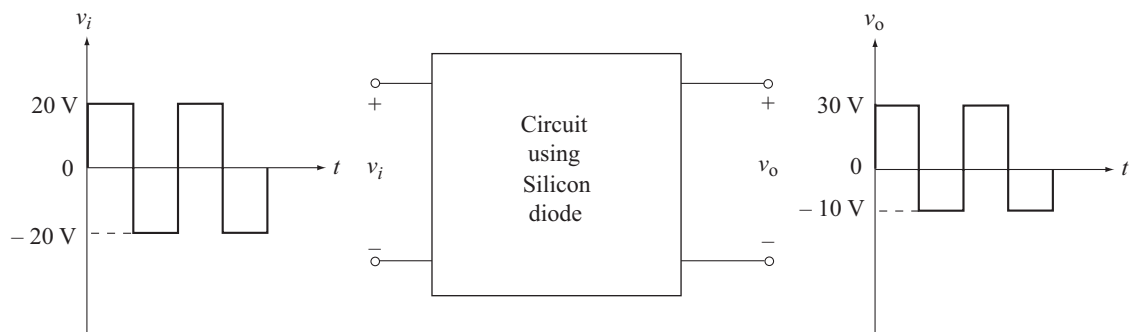
Fig. 1.50 Output voltage waveform
(a) With practical diode
(b) With ideal diode

Since the circuit adds positive dc level to the input signal, it is called positive clamper. The circuit is also called negative peak clamper, since the negative peak of the applied signal is clamped at 0 V as shown in Fig. 1.50. Once again we observe from Fig. 1.50 that, the peak-to-peak output voltage equals the peak-to-peak input voltage.

If clamping is desired at a voltage other than zero, a reference voltage V_R must be included in series with diode. Such circuits are considered later in numerical examples.

Example 1.52

Design a suitable circuit represented by the box shown below which has input and output waveforms as indicated.

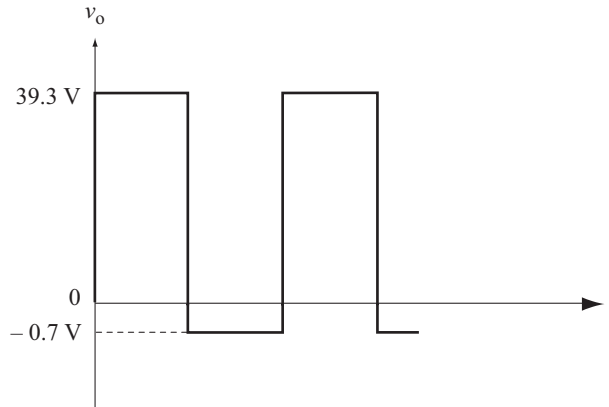


Solution

Comparing the waveforms of v_i and v_o , we find that, v_o can be obtained by shifting up v_i by 10 V. To do so, a positive dc voltage must be added to v_i . Hence we have to design a positive clamper circuit.

Let us sketch the output of positive clamper circuit of Fig. 1.49 for the given input voltage. (Refer Table 1.7)

$$V_m = 20 \text{ V} \quad V_K = 0.7 \text{ V}$$

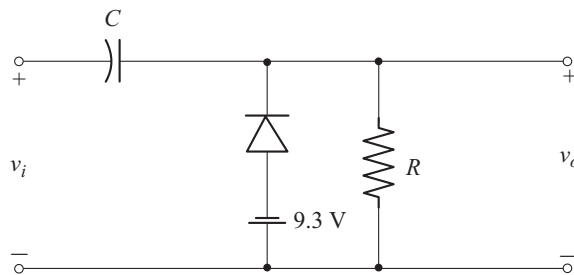


But the desired output voltage has a positive peak of 30 V. Therefore, the waveform shown above has to be shifted down by 9.3 V. This can be done by using an additional dc source V_R , whose value is given by

$$39.3 \text{ V} + V_R = 30 \text{ V}$$

$$V_R = -9.3 \text{ V}$$

This dc voltage should be placed in series with the diode. The required clamper circuit is shown below.

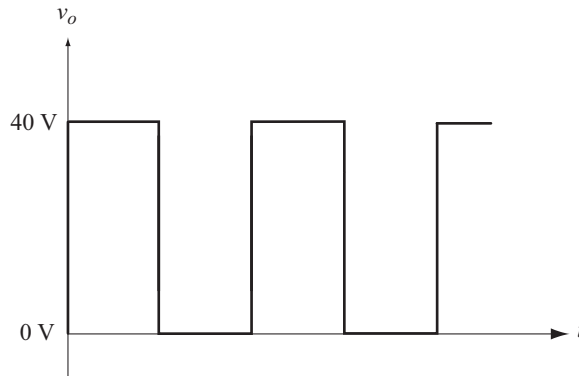


Example 1.53

Repeat Example 1.52 using ideal diode.

$$V_m = 20 \text{ V} \quad V_K = 0.$$

The output of positive clamper is shown below.

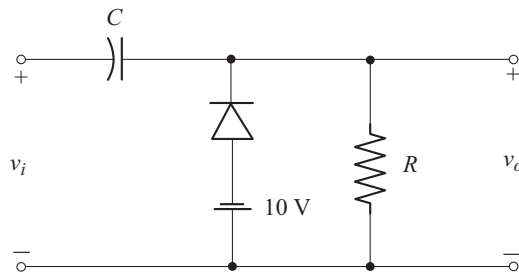


Since the required positive peak is 30 V,

$$40 \text{ V} + V_R = 30 \text{ V}$$

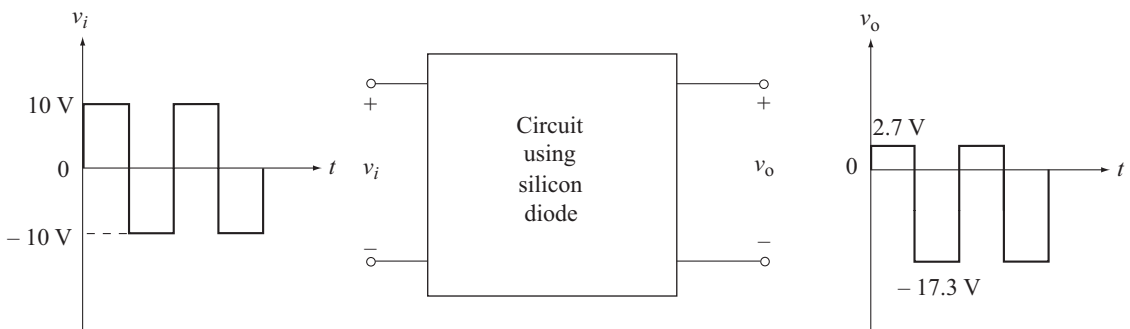
$$V_R = -10 \text{ V}$$

The clamper circuit is shown below.



Example 1.54

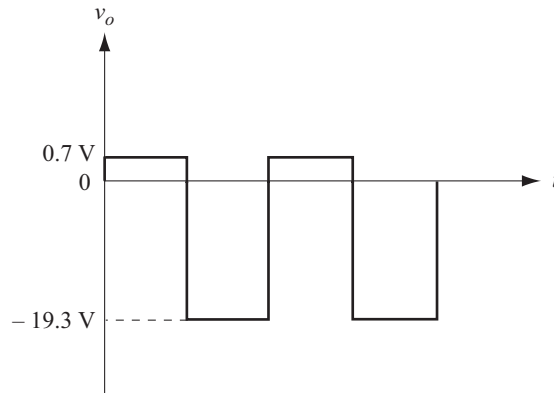
Design a suitable circuit represented by the box shown below which has the input and output waveforms as indicated.



Solution

Comparing the waveforms of v_i and v_o , we find that, v_o can be obtained by shifting down v_i by 7.3 V. To do so, a negative dc voltage must be added to v_i . Hence we have to design a negative clamper. The output of negative clamper of Fig. 1.46 for the given input is shown below. (Refer Table 1.6).

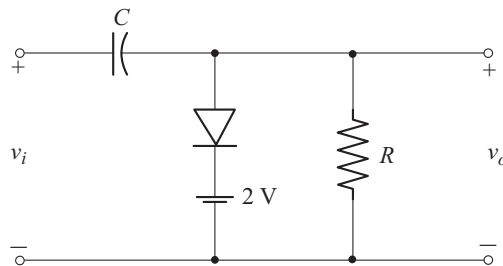
$$V_m = 10 \text{ V} \quad V_K = 0.7 \text{ V}$$



The desired output voltage has a positive peak of 2.7 V. Therefore the waveform shown above has to be shifted up by 2 V. This can be done by using an additional dc source V_R given by

$$\begin{aligned} 0.7 \text{ V} + V_R &= 2.7 \text{ V} \\ V_R &= 2 \text{ V} \end{aligned}$$

This dc voltage should be placed in series with the diode. The required clamper circuit is shown below.

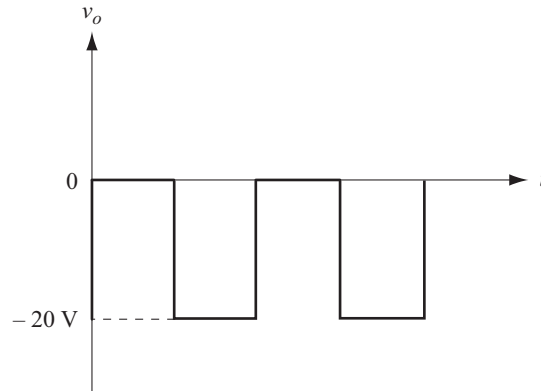
**Example 1.55**

Repeat Example 1.54 using ideal diode.

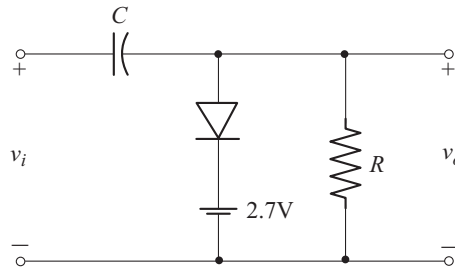
Solution

$$V_m = 10 \text{ V} \quad V_K = 0$$

The output of negative clamper is shown below.

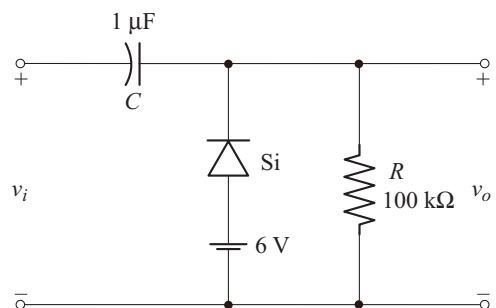
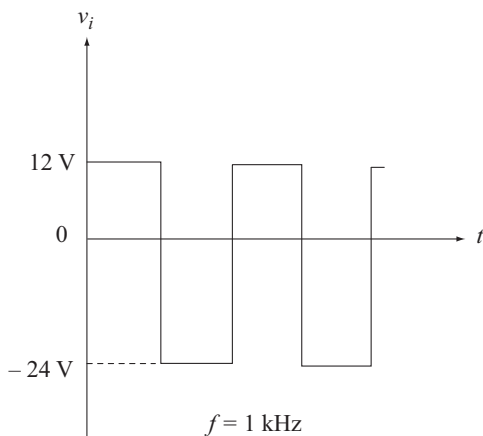


Since the required positive peak is 2.7 V, the waveform shown should be shifted up by 2.7 V. Therefore a dc source of voltage 2.7 V must be placed in series with the diode as shown below.



Example 1.56

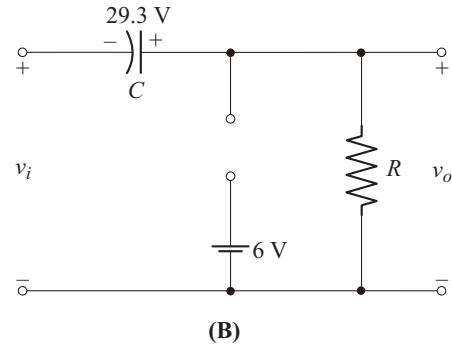
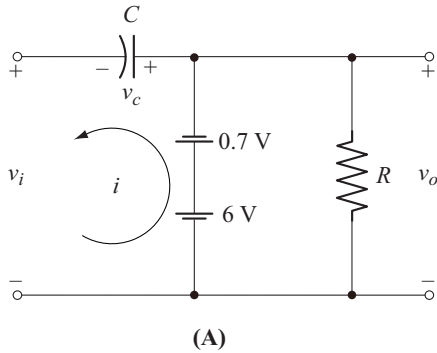
For the circuit shown below find and plot the waveform of v_o for the input indicated.



Solution

Step 1: To find the capacitor voltage when the diode is conducting.

Diode conducts when $v_i = -24$ V. The equivalent circuit is shown in Fig. A.



Applying KVL to the current path in the circuit of Fig. A we have

$$v_i + v_c + 0.7 \text{ V} - 6 \text{ V} = 0 \quad (\text{A})$$

$$v_c = -v_i + 5.3 \text{ V}$$

When $v_i = -24 \text{ V}$, the capacitor charges to

$$v_c = -(-24 \text{ V}) + 5.3 \text{ V} = 29.3 \text{ V}$$

Note that the right plate of capacitor is positive.

Step 2: To find the output voltage levels when the diode is off

When $v_i = 12 \text{ V}$, the diode is off. The equivalent circuit is shown in Fig. B.

KVL equation to the circuit of Fig. B is

$$v_i + 29.3 \text{ V} - v_o = 0 \quad (\text{B})$$

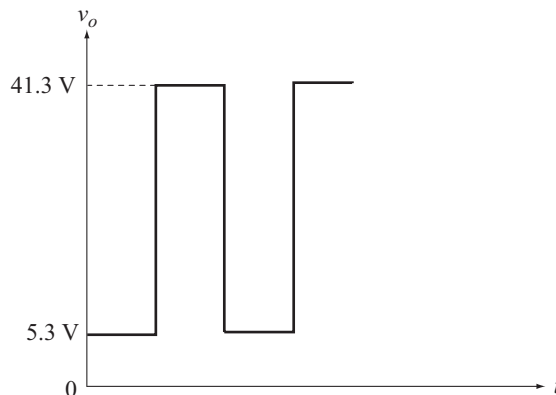
$$v_o = v_i + 29.3 \text{ V}$$

Observe from Equation (B) that, the circuit adds a dc level of 29.3 V to the input signal v_i . Let us find the output levels corresponding to the two input levels using Equation (B).

$$\text{When } v_i = 12 \text{ V}, \quad v_o = 12 \text{ V} + 29.3 \text{ V} = 41.3 \text{ V}$$

$$\text{When } v_i = -24 \text{ V}, \quad v_o = -24 \text{ V} + 29.3 \text{ V} = 5.3 \text{ V}$$

The output waveform is shown below.



$$\text{Peak-to-peak input voltage} = 12 \text{ V} - (-24 \text{ V}) = 36 \text{ V}$$

$$\text{Peak-to-peak output voltage} = 41.3 \text{ V} - 5.3 \text{ V} = 36 \text{ V}$$

Note that peak-to-peak output voltage = peak-to-peak input voltage.

Example 1.57

Repeat example 1.56 assuming ideal diode.

Solution

For ideal diode, $V_K = 0$

From Equation (A),

$$v_c = -v_i + 6 \text{ V}$$

When $v_i = -24 \text{ V}$

$$v_c = -(-24 \text{ V}) + 6 \text{ V} = 30 \text{ V}$$

From Equation (B),

$$v_o = v_i + 30 \text{ V}$$

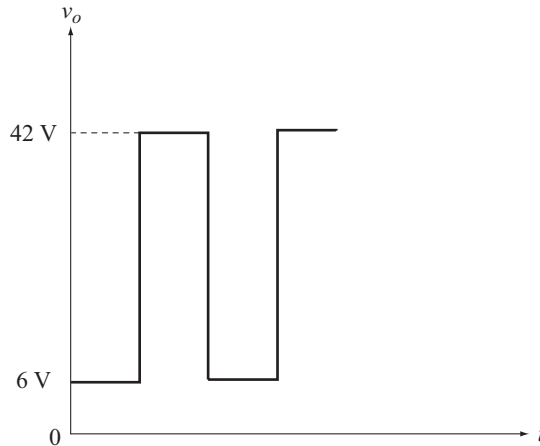
When $v_i = 12 \text{ V}$,

$$v_o = 12 \text{ V} + 30 \text{ V} = 42 \text{ V}$$

When $v_i = -24 \text{ V}$,

$$v_o = -24 \text{ V} + 30 \text{ V} = 6 \text{ V}$$

The output waveform is shown below.

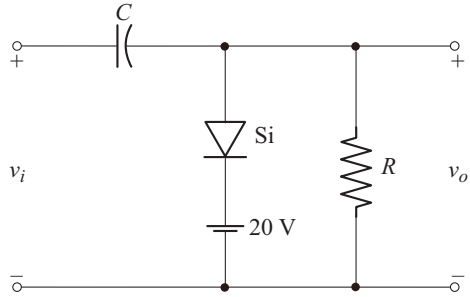
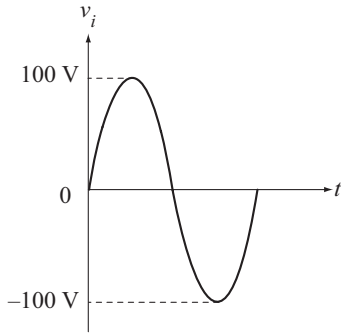


Once again we note that

$$\text{Output peak-to-peak} = \text{input peak-to-peak} = 36 \text{ V}$$

Example 1.58

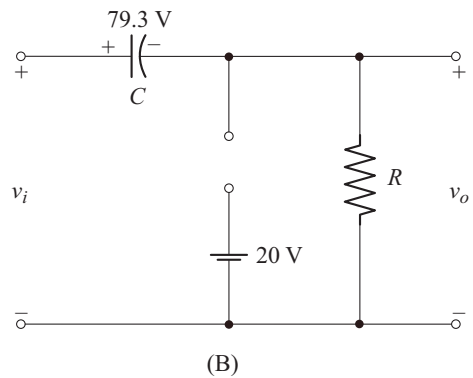
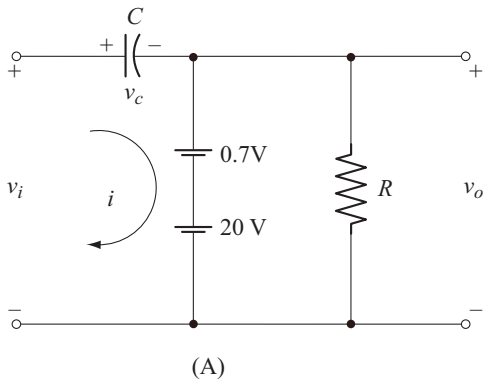
For the circuit shown find and plot the output waveform for the input indicated. Also sketch the output waveform assuming ideal diode.



Solution

Step 1 : To find capacitor voltage when diode is conducting

Diode conducts during the positive half cycle of v_i . The equivalent circuit is shown in Fig. A.



Applying KVL to the current path in the circuit of Fig. A.

$$v_i - v_c - 0.7 \text{ V} - 20 \text{ V} = 0 \quad (\text{A})$$

$$v_c = v_i - 20.7 \text{ V}$$

When $v_i = 100 \text{ V}$, the capacitor charges to

$$v_c = 100 \text{ V} - 20.7 \text{ V} = 79.3 \text{ V}$$

Step 2 : To find the output voltage levels when the diode is off

Diode turns off during the negative half cycle of v_i . The equivalent circuit is shown in Fig. B.

KVL equation to the circuit of Fig. B is

$$v_i - 79.3 \text{ V} - v_o = 0$$

$$v_o = v_i - 79.3 \text{ V} \quad (\text{B})$$

$$\text{When } v_i = 100 \text{ V,}$$

$$v_o = 100 \text{ V} - 79.3 \text{ V} = 20.7 \text{ V}$$

$$\text{When } v_i = -100 \text{ V,}$$

$$v_o = -100 \text{ V} - 79.3 \text{ V} = -179.3 \text{ V}$$

Output Waveform with Ideal Diode

From Equation (A),

$$v_c = v_i - 20 \text{ V}$$

$$\text{When } v_i = 100 \text{ V,}$$

$$v_c = 100 \text{ V} - 20 \text{ V} = 80 \text{ V}$$

From Equation (B)

$$v_o = v_i - 80 \text{ V}$$

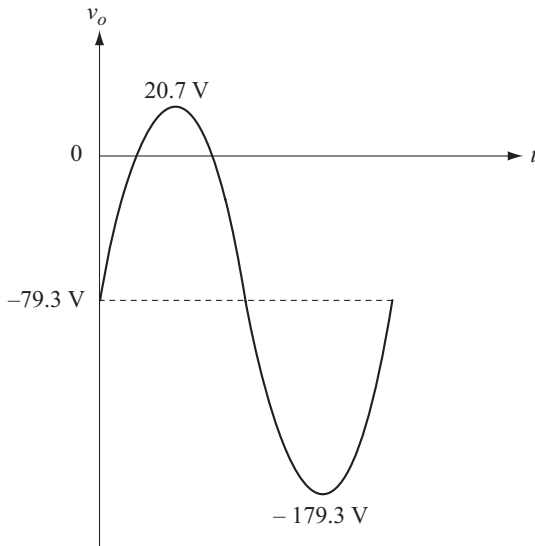
$$\text{When } v_i = 100 \text{ V,}$$

$$v_o = 100 \text{ V} - 80 \text{ V} = 20 \text{ V}$$

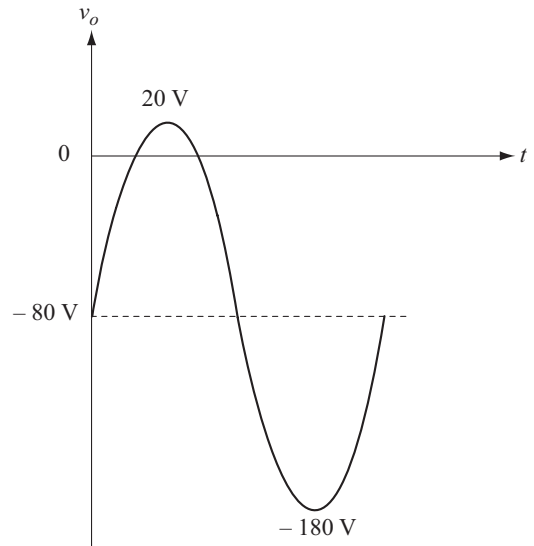
$$\text{When } v_i = -100 \text{ V,}$$

$$v_o = -100 \text{ V} - 80 \text{ V} = -180 \text{ V}$$

The output waveforms are shown below.



Output waveform with Si diode



Output waveform with ideal diode

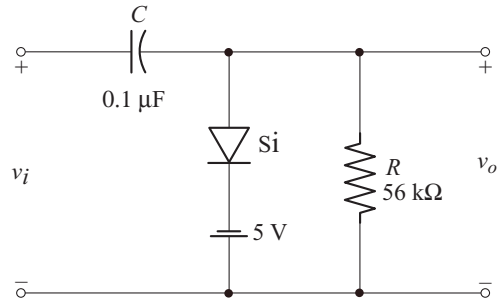
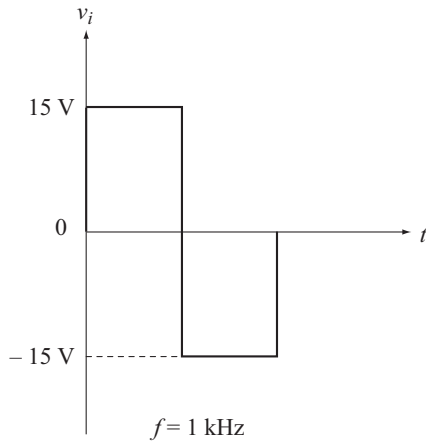
Note that in each case

$$\text{output peak-to-peak} = \text{input peak-to-peak} = 200 \text{ V}$$

Example 1.59

For the circuit shown below

- Calculate 5τ
- Compare 5τ with half period ($T/2$) of the applied signal
- Sketch the output waveform

**Solution**

$$(a) \tau = RC = (56 \text{ k}\Omega)(0.1 \text{ }\mu\text{F}) = 5.6 \text{ ms}$$

$$5\tau = 5(5.6 \text{ ms}) = 28 \text{ ms}$$

$$(b) f = 1 \text{ kHz} \quad T = \frac{1}{f}$$

$$T = \frac{1}{1 \text{ kHz}} = 1 \text{ ms} \Rightarrow \frac{T}{2} = 0.5 \text{ ms}$$

$$\frac{5\tau}{\left[\frac{T}{2}\right]} = \frac{28 \text{ ms}}{0.5 \text{ ms}} = 56$$

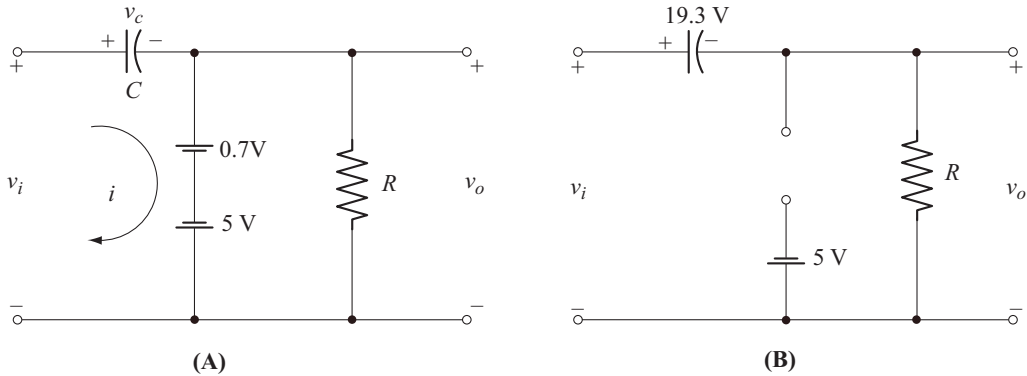
$$\therefore 5\tau = 56 \frac{T}{2} \Rightarrow 5\tau \gg \frac{T}{2}$$

Due to this large time constant the discharge of the capacitor is very minimal when the diode is off.

(c) To find and plot v_o

Step 1: To find the capacitor voltage when the diode is conducting

Diode conducts when $v_i = 15 \text{ V}$. The equivalent circuit is shown in Fig. A.



Applying KVL to the current path in the circuit of Fig. A, we have

$$v_i - v_c - 0.7\text{ V} + 5\text{ V} = 0$$

$$v_c = v_i + 4.3\text{ V}$$

When $v_i = 15\text{ V}$, the capacitor charges to

$$v_c = 15\text{ V} + 4.3\text{ V} = 19.3\text{ V}$$

Step 2: To find output voltage levels when the diode is off

Diode turns off when $v_i = -15\text{ V}$. The equivalent circuit is shown in Fig. B.

KVL equation to the circuit of Fig. B is

$$v_i - 19.3\text{ V} - v_o = 0$$

$$v_o = v_i - 19.3\text{ V}$$

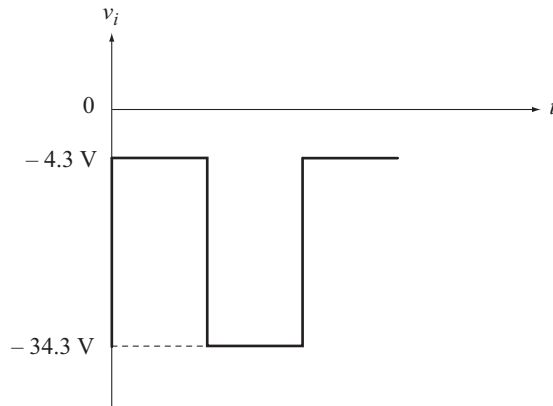
When $v_i = 15\text{ V}$,

$$v_o = 15\text{ V} - 19.3\text{ V} = -4.3\text{ V}$$

When $v_i = -15\text{ V}$,

$$v_o = -15\text{ V} - 19.3\text{ V} = -34.3\text{ V}$$

The output waveform is shown below.



Note that, output peak-to-peak = input peak-to-peak = 30 V

**Exercise Problems**

- 1.1 A full-wave bridge rectifier using silicon diodes is supplying a resistive load of $1\text{ k}\Omega$. It is supplied from an ac source of 100 V at 50 Hz . Calculate
 - (a) DC output voltage
 - (b) DC load current
- 1.2 The input to a clipping circuit is a sine wave of peak value 25 V . Design the component values such that the output should have its positive peak clipped at 15 V and negative peak at -18 V .
- 1.3 Design the values of R and C for a clamping circuit for which the input signal is a square wave of frequency 1 kHz .
- 1.4 Design a clamping circuit to obtain an output with positive peak at 20 V and negative peak at -10 V . The input is a square wave of $\pm 15\text{ V}$ at 1 kHz . Assume silicon diode.
- 1.5 A shunt clipper employs a diode which has $r_f = 100\ \Omega$ and $R_r = 100\text{ k}\Omega$. Calculate the reasonable value of R .

Chapter 2

TRANSISTOR BIASING

Biasing a bipolar junction transistor essentially means establishing the desired value of collector-emitter voltage V_{CE} and the collector current I_C , to ensure that the amplifier will have the proper gain and input impedance with undistorted output voltage swing. These values of V_{CE} and I_C are together known as the *quiescent operating point* or *Q-point*.

Now, one of the basic problems with the transistor amplifier is establishing and maintaining proper values of quiescent V_{CE} and I_C in the circuit. A quiescent voltage or current refers to the values under dc conditions in the absence of any ac input signal. The quiescent value of voltages and currents are maintained by using configurations that assure stability against variations in temperature. Variations in temperature affects several critical transistor parameters such as reverse saturation current I_{CO} , base-emitter voltage in the active region V_{BE} and current gain β .

This chapter analyses various bias configurations and techniques for maintaining the quiescent operating point stable against variations in temperature. The concepts are introduced with a generous number of illustrative examples and in-depth analysis of different configurations.

◆ 2.1 THE OPERATING POINT

The transistor needs to be operated in an appropriate region of its characteristics depending on the application of the circuit in which it is being used. The dc currents and voltages in the circuit are established by using a resistive network along with a dc power supply. This process is called *biasing*.

Let us recollect the common-emitter output characteristics of an *n-p-n* transistor. The common emitter configuration and its output characteristics are shown in Fig. 2.1.

The three regions of operation of the transistor are termed as active region, cut-off region and saturation region.

The region in the characteristics shown in Fig. 2.1(b) above $I_B = 0$ mA and to the right of a few tenths of volts of V_{CE} is the active region. It is the region where you observe an increase in the collector current with increase in base current. The base-emitter junction is forward biased while the base-collector junction is reverse biased in this region.

The region where the base-emitter junction is reverse biased below 0.1 V for germanium and 0 V for silicon is the cut-off region. In this region, the emitter current is zero and the transistor is non-conducting. In this region the base-emitter junction as well as the base-collector junction is reverse biased.

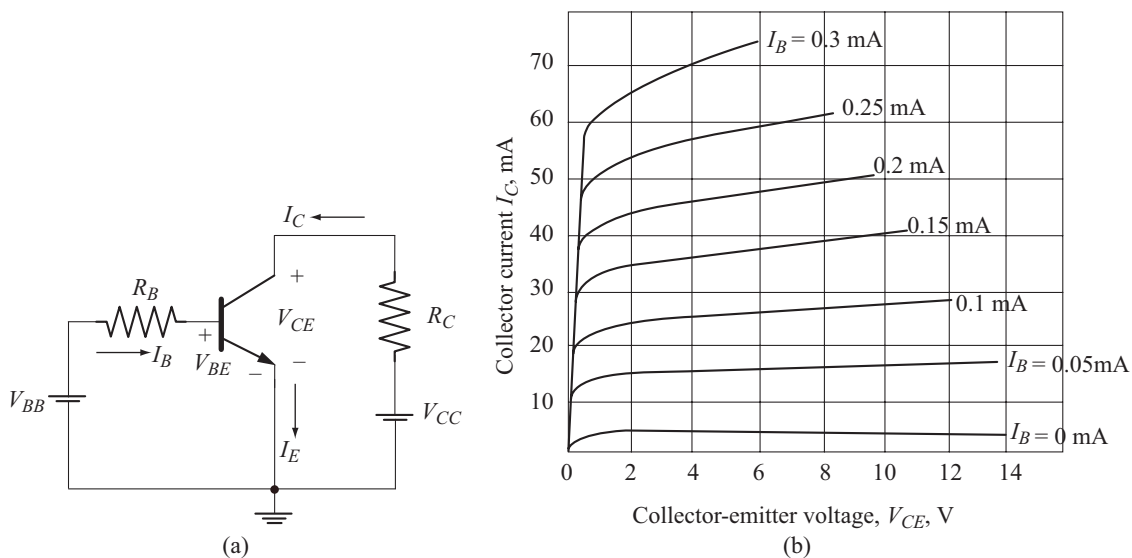


Fig. 2.1 The common-emitter configuration and its output characteristics

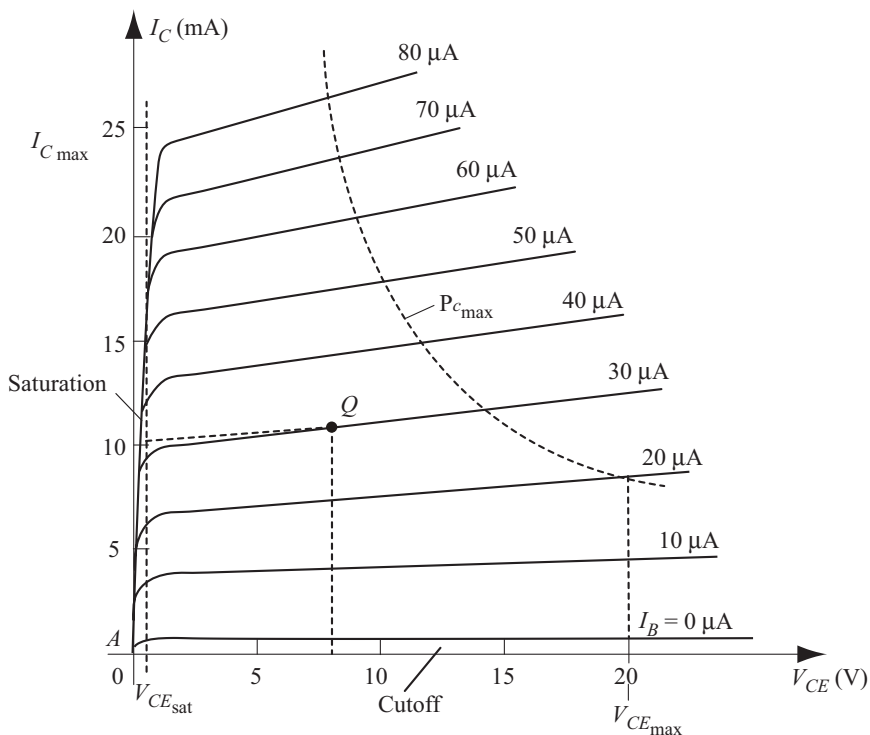


Fig. 2.2 Illustration of operating point

The region very close to $V_{CE} = 0$ where all the curves appear to merge and fall rapidly to the origin is called the *saturation region*. In this region, V_{CE} is a few tenths of volts while I_C is considerably large. In the saturation region the base-emitter and the base-collector junctions are forward biased.

The transistor is required to be biased from cut-off to saturation and vice-versa when it is being used as a switch. It must be biased in the active region when being used as an amplifier. The transistor functions linearly when its operation is restricted to the active region. For transistor amplifiers, we need to establish an operating point on the characteristics to define a region of operation by fixing the dc current and voltages.

The operating point of a transistor for $I_C = 10$ mA and $V_{CE} = 8$ V is marked as Q in the characteristics shown in Fig. 2.2.

Since, the operating point is a fixed point on the characteristics, it is also referred to as the quiescent point or Q -point. The Q -point can be fixed anywhere in the active region bounded as shown in Fig. 2.2 and as tabulated in Table 2.1.

Table 2.1 Boundaries of the active region

<i>Boundary</i>	<i>Limiting factor</i>
Left bound	Saturation region
Top Bound	$I_{C \text{ max}}$
Bottom bound	Cut-off region
Right bound	$P_{C \text{ max}}$

$P_{C \text{ max}}$ is the maximum power dissipation which the device can withstand.

◆ 2.2 DC LOAD LINE

As in the case of a diode, a line called the dc load line can be drawn on the characteristics of the transistor also, which represents the applied load. The intersection of the load line with the characteristics will determine the operating point. Let us now see how to draw the dc load line.

Consider the collector circuit of a biased transistor shown along with its characteristics in Fig. 2.3.

The KVL Equation for the collector circuit is

$$\therefore V_{CC} = I_C R_C + V_{CE} \quad (2.1)$$

The V_{CE} axis intercept can be found by letting $I_C = 0$ in Equation (2.1).

$\therefore V_{CE} = V_{CC}$ and the intercept is at $(V_{CC}, 0)$. The I_C axis intercept can be found by letting $V_{CE} = 0$ in Equation (2.1).

$$\therefore I_C = \frac{V_{CC}}{R_C} \text{ and the intercept is at } \frac{V_{CC}}{R_C}, 0$$

The load line is plotted by joining these two intercepts as shown in Fig. 2.3(b).

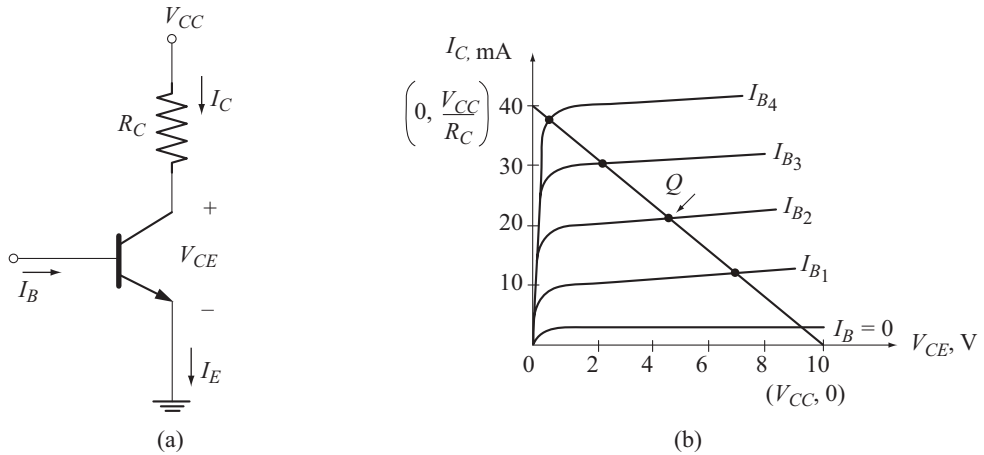


Fig. 2.3 Biased transistor along with its characteristics showing the dc load line

The equation for the dc load line in the slope intercept form can be written from Equation (2.1) as

$$I_C = -\frac{1}{R_C} V_{CE} + \frac{V_{CC}}{R_C} \quad (2.2)$$

where the slope is $-\frac{1}{R_C}$ while the intercept is $\frac{V_{CC}}{R_C}$. The intersection of this load line with the output characteristics would result in a number of possible operating points. However, the operating point is chosen around the middle of the load line to provide an equal swing of I_C and V_{CE} about the point. This would ensure that the transistor remains in the active region during its entire operation and not stray temporarily into the cut-off or saturation regions. The coordinates of an optimum operating point is therefore $\frac{V_{CC}}{2}, \frac{V_{CC}}{2R_C}$.

◆ 2.3 REASONS FOR INSTABILITY OF THE OPERATING POINT

Having selected and biased the transistor at a desired operating point, the effect of temperature on various device parameters can cause this operating point to drift.

Let us now look at the effects of temperature on transistor parameters. The reasons for instability of the operating point may be listed as

- Variation of leakage current, I_{CO} with temperature
- Variation of current gain, β with temperature and
- Variation of base-emitter voltage, V_{BE} with temperature.

The sensitivity of these parameters to temperature variations can be stated as follows:

- I_{CO} , the reverse saturation current doubles for every 10°C increase in temperature
- V_{BE} decreases in magnitude by 2.5 mV for every $^\circ\text{C}$ rise in temperature. Generally at room temperature V_{BE} is 0.7 V for silicon transistors and 0.2 V for germanium transistors.
- β increases with rise in temperature.